

Adult-born granule cells modulate CA2 network activity during retrieval of developmental memories of the mother

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Blake J. Laham, Isha R. Gore, Casey J. Brown, Elizabeth Gould 

Princeton Neuroscience Institute, Princeton University, Princeton NJ 08544

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Abstract

Adult-born granule cells (abGCs) project to the CA2 region of the hippocampus, but it remains unknown how this circuit affects behavioral function. Here we show that abGC input to the CA2 of adult mice is involved in the retrieval of remote developmental memories of the mother. Ablation of abGCs impaired the ability to discriminate between a caregiving mother and a novel mother, and this ability returned after new neurons were regenerated. Chemogenetic inhibition of projections from abGCs to the CA2 also temporarily prevented the retrieval of remote mother memories. These findings were observed when abGCs were inhibited at 4-6 weeks old, but not when they were inhibited at 10-12 weeks old. We also found that abGCs are necessary for differentiating features of CA2 network activity, including theta-gamma coupling and sharp wave-ripples, in response to novel versus familiar social stimuli. Taken together, these findings suggest that abGCs are necessary for neuronal oscillations associated with discriminating between social stimuli, thus enabling retrieval of remote developmental memories of the mother by their adult offspring.

eLife assessment

This paper reports a **valuable** set of new results. The main result is that the projection from adult-born granule cells in the dentate gyrus to the hippocampal subfield CA2 is necessary for the retrieval of a social memory formed during development, and **solid** evidence is provided to support this conclusion.

Introduction

The earliest social memories arise as infants learn to recognize their first caregiver - most commonly, the mother. In humans, these early associations can be further strengthened by emotional valence and shared experiences as the child develops, forming memories of the mother that can last a lifetime (Fivush, 2011 [↗](#); Schaal et al., 2020 [↗](#)). Mice can also recognize their mothers and distinguish them from unfamiliar mothers during the first postnatal week (Laham et al., 2021 [↗](#)). These memories last well into adulthood, with a reversal in social preference at the

time of weaning (Laham et al., 2021 [DOI](#)); pup offspring prefer investigating their own mothers over novel mothers, while adult offspring prefer investigating unfamiliar mothers over their own. In adulthood, the ability to discriminate between the caregiving mother and a novel mother persists despite any contact with the mother for over a month after weaning. Lower investigation times of the mother compared to a novel mother suggest retrieval of a remote social memory by adult offspring.

The CA2 region of the hippocampus is required for the ability to recognize caregiving mothers and distinguish them from novel mothers both during development and in adulthood (Laham et al., 2021 [DOI](#)). The CA2 is also necessary for short-term memory of peers during development and in adulthood (Hitti and Siegelbaum, 2014 [DOI](#); Smith et al., 2016 [DOI](#); Laham et al., 2021 [DOI](#); Diethorn and Gould, 2023 [DOI](#)). A growing literature suggests that rhythmic firing in the CA2 supports the social memory functions of this region. Phase-amplitude coupling (PAC) of low frequency theta rhythm with higher frequency gamma rhythm in the CA2 and the generation of sharp wave-ripples (SWRs), high frequency bursts in the CA2, are increased by exposure to familiar peers and short-term consolidation of social memory for peers, respectively (Zhu et al., 2022 [DOI](#); Oliva et al., 2020 [DOI](#)). In other hippocampal subregions, namely the CA1 and CA3, PAC and SWRs have been associated with novelty detection, memory consolidation, and memory retrieval of nonsocial stimuli and events over relatively short time frames (Colgin, 2015 [DOI](#); Fernández-Ruiz et al., 2019 [DOI](#); Joo and Frank, 2018 [DOI](#); Meier et al., 2020 [DOI](#); Vivekananda et al., 2021 [DOI](#)). These studies suggest that certain patterns of oscillatory firing may facilitate replay and pre-play of neuronal ensembles activated during specific experiences (Joo and Frank, 2018 [DOI](#)). However, no previous studies have examined whether PAC and SWRs are associated with retrieval of remote developmental memories.

The CA2 has been called a social memory “hub” because it receives and integrates inputs from multiple brain areas (Hitti and Siegelbaum, 2014 [DOI](#); Oliva et al., 2020 [DOI](#); Diethorn and Gould, 2023 [DOI](#)), including from granule cells of the dentate gyrus (Kohara et al., 2014 [DOI](#)), a population known to undergo adult neurogenesis (Song et al., 2012 [DOI](#)). Adult-born granule cells (abGCs) are important for discriminating between novel and familiar peers (Pereira-Caixeta et al., 2018 [DOI](#); Cope et al., 2020 [DOI](#)), and although these cells are known to project to the CA2 (Llorens-Martin et al., 2015), the function of this circuit remains unexplored. Furthermore, although the dentate gyrus is known to influence hippocampal gamma oscillations, PAC, and SWRs in general (Bott et al., 2016 [DOI](#); Meier et al., 2020 [DOI](#); Fernández-Ruiz et al., 2021 [DOI](#)), no studies have examined the influence of abGCs on these oscillatory patterns in the CA2 specifically. Here we investigated whether abGCs support the retrieval of developmental memories of the mother, and whether these cells contribute to socially relevant CA2 network activity.

Results

abGCs project to inhibitory interneurons and pyramidal cells of the CA2 region

In the CA2, we verified abGC projections using 3R-Tau, a marker of abGC axons (Llorens-Martin et al., 2015), as well as mature granule cell projections using ZnT3, a marker of mature mossy fibers (Wenzel et al., 1997 [DOI](#)) (Figure 1a [DOI](#), ai-aiii). We further determined that abGC projections to the CA2 are preferentially associated with the proximal dendrites of PV+ interneurons labeled with cre-dependent AAV mCherry compared to the proximal dendrites of pyramidal cells immunolabeled with PCP4 (Figure 1b-d [DOI](#)). Although we cannot determine whether a similar relationship exists with projections to distal dendrites or whether these connections represent active synapses, this preferential association with PV+ interneurons over pyramidal cells mirrors abGC projection patterns to the CA3 (Restivo et al., 2015 [DOI](#)). Because the CA2 region is involved in the retrieval of remote developmental memories (Laham et al., 2021 [DOI](#)) and abGCs have been

linked to the storage and retrieval of recently encountered mice (Cope et al., 2020 [↗](#)), we next investigated whether abGCs are necessary for the retrieval of developmental memories of the mother.

abGCs are necessary for retrieval, but not maintenance, of remote developmental memories

Using the GFAP-TK transgenic mouse model in which abGCs are ablated upon valganciclovir (VGCV) administration (Snyder et al., 2011 [↗](#)), we found that adult GFAP-TK mice and CD1 littermates with control levels of abGCs investigate their own mother less than a novel mother (Figure 1e-g [↗](#); 1 1a), similar to what has been observed in C57 mice (Laham et al., 2021 [↗](#)). Because adult mice prefer novel stimuli over familiar stimuli, lower investigation times of previously encountered mice compared to novel mice can be interpreted as familiarity recognition. Thus, our social investigation data in adult mice with intact abGCs suggests that these mice can recognize mothers as familiar even after being separated since weaning (at P21). In contrast, mice with depleted abGCs are no longer able to discriminate between their own mother and a novel mother, as indicated by investigation times (Figure 1g [↗](#); Supplementary Figure 1a [↗](#)). No changes were observed in overall locomotion during social trials after abGC depletion (Supplementary Figure 2a [↗](#)), suggesting that any physical activity differences between groups are not responsible for differences in social investigation times. Cessation of VGCV treatment in GFAP-TK mice enabled a 70% recovery of abGCs in the DG (Figure 1h [↗](#), j, k) and complete recovery of abGC 3R-Tau+ axon intensity in the CA2 (Figure 1i [↗](#), l, m). Coinciding with the recovery of adult neurogenesis, GFAP-TK animals regained the ability to discriminate between their mother and a novel mother with differential investigation times (Figure 1g [↗](#), Supplementary Figure 1a [↗](#)). These results reveal that abGCs are not necessary for maintaining developmental social memories, but instead support retrieval of remote memories of the mother. Because it is known that abGCs project to the CA2 (Llorens-Martin et al., 2015), a region which has been linked to retrieval of developmental social memories (Laham et al., 2021 [↗](#)), we next inhibited projections from abGCs to the CA2 in order to investigate how this circuit affects mother memory retrieval.

abGC projections to the CA2 play a time-limited role in the retrieval of remote developmental memories

To determine if developmental social memory retrieval requires activation of the abGC-CA2 circuit, we used a tamoxifen-inducible double transgenic mouse where Gi-DREADD expression was confined to abGCs in a temporally specific way (Nestin-cre:Gi). Nestin-cre littermates were used as single transgenic controls. Targeted clozapine-N-oxide (CNO) infusion through bilateral cannulae aimed at the CA2 enabled inhibition of Gi-DREADD+ abGC axon terminals in the vicinity of the cannulae in Nestin-cre:Gi mice. Previous studies have shown that abGCs are important for nonsocial memories only during a limited time after they are generated, i.e., 4-6 weeks after mitosis (Gu et al., 2012 [↗](#); Song et al., 2012 [↗](#); Kropff et al., 2015 [↗](#)). Accordingly, tamoxifen-treated adult mice underwent behavioral testing at two post-injection time points (4-6 weeks and 10-12 weeks post injection) to assess whether different aged abGCs are involved in developmental social memory retrieval (Figure 2a-d [↗](#)). As expected, we found that Nestin-cre:Gi mice investigated the mother less than the novel mother when vehicle was infused; however inhibition of this pathway after CNO infusion in CA2 prevented this ability (Figure 2e [↗](#), Supplementary Figure 1b [↗](#)). There were no differences in locomotion at baseline or during the social trials in any groups, suggesting that the effects of social investigation time were not the result of physical activity differences (Supplementary Figure 2b, c [↗](#)). CNO treatment in the same abGC cohort 6 weeks later (10-12-week-old abGCs) had no effect on Nestin-cre:Gi animals' ability to discriminate between their mother and a novel mother (Figure 2f [↗](#), Supplementary Figure 1c [↗](#)). These findings suggest that abGCs are required for retrieving remote social memories of the mother

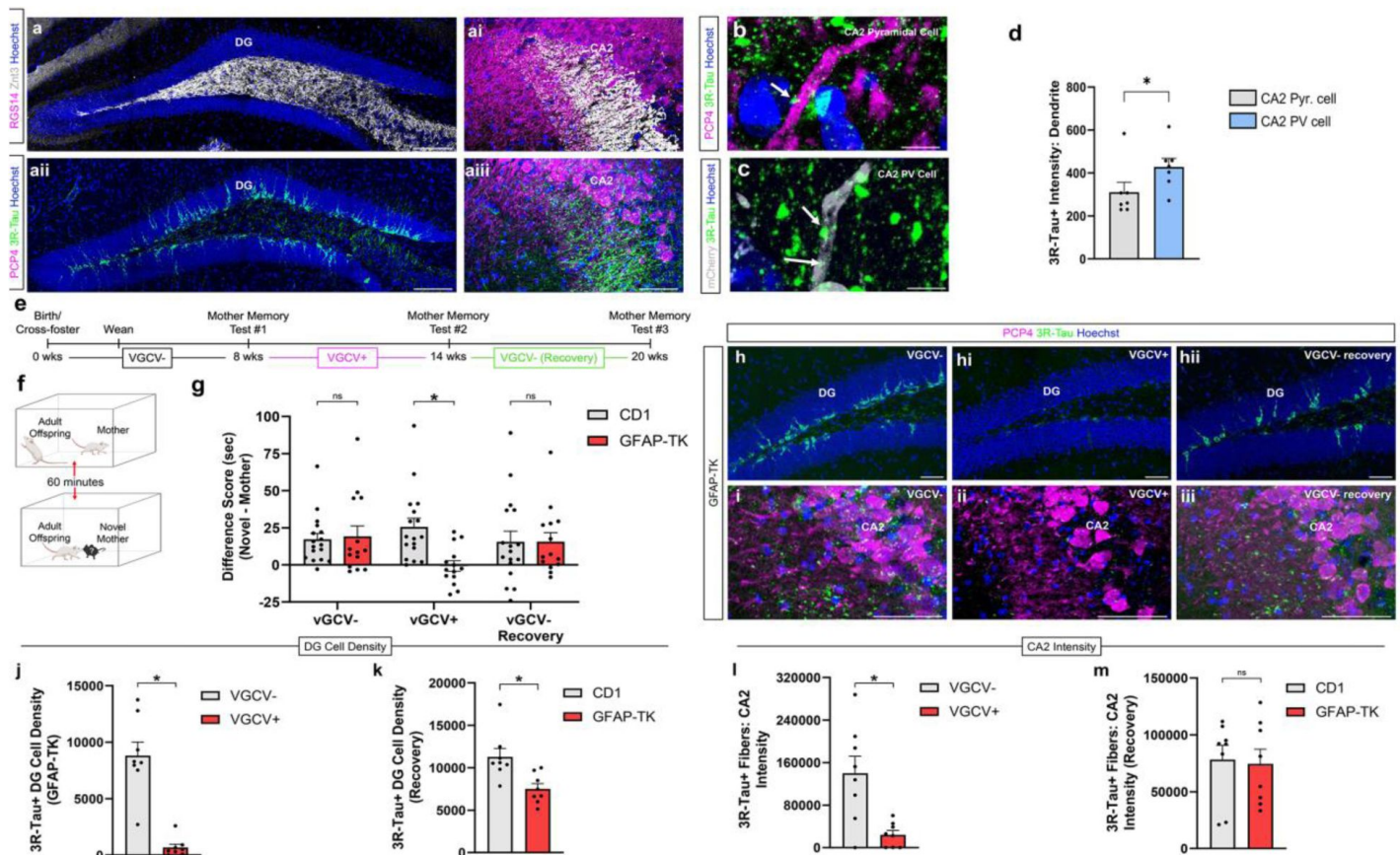


Figure 1

Adult-born granule cells support retrieval of developmental remote memories of the mother.

(**a-aiii**) Confocal images of dentate gyrus granule cells, both mature (ZnT3+) and adult-born (3R-Tau+), and their robust projections to CA2. Confocal images of abGC 3R-Tau+ axons (arrows) near proximal dendrite of a CA2 PCP4+ pyramidal cell (**b**) and near proximal dendrite of a CA2 mCherry+ PV+ interneuron (**c**). **d** abGC fibers (3R-Tau+) in the CA2 are more abundant near proximal dendrites ($n = 7$; paired t test: $t_6 = 3.163$, $p = 0.0195$) of PV+ interneurons (mCherry+) than those of pyramidal cells (PCP4+). **e** Timeline for behavioral experiment. **f** Schematic demonstrating direct social interaction assay used at 3 experimental time points. **g** Ablation of abGCs abolishes difference in investigation time between mother and novel mother (CD1: $n = 17$; TK: $n = 14$; 2-way RM ANOVA: Genotype $\times$ Drug: $F_{2,58} = 5.352$, $p = 0.0074$; Šidák's test $p = 0.0021$). After removing VGCV from rodent chow and allowing adult neurogenesis to recovery over the course of 6 weeks, mice were able to discriminate between mother and novel mother again ($p > 0.999$). (**h-iii**) Confocal images of abGCs in the DG at 3 different drug time points. VGCV administration produces a dramatic loss of (**h-iii**) abGCs (3R-Tau+ cells) in DG (VGCV-: $n = 8$; VGCV+: $n = 8$; unpaired t test: $t_{14} = 6.588$, $p < 0.0001$) and (**i-iii**) 3R-Tau+ fibers in CA2 (VGCV-: $n = 8$; VGCV+: $n = 8$; unpaired t test: $t_{14} = 3.488$, $p = 0.0036$). 6 weeks after removing VGCV from the rodent chow, DG abGC number (**j,k**) undergoes a 70% recovery (CD1: $n = 8$; TK: $n = 8$; unpaired t test: $t_{14} = 3.218$, $p = 0.0062$), and (**l,m**) 3R-Tau+ fibers present in CA2 undergo a complete recovery (CD1: $n = 8$; TK: $n = 8$; unpaired t test: $t_{14} = 0.7271$, $p = 0.4791$). $*p < 0.05$, bars represent mean $\pm$ SEM. Scale bars = 200 μ m for **a, h, i**; 2 μ m for **b,c**. ns = not significant

when the cells are immature (4-6 weeks post-mitosis), but not when they are mature (10-12 weeks post-mitosis), presumably because a younger abGC cohort has taken over this function. It should be noted that although our cannulae were placed over the CA2 region, we cannot rule out the possibility that CNO affected abGC terminals in the nearby CA3a region. An additional experiment explored if abGCs contribute to peer social memory consolidation (**Figure 2g**). Immature abGCs were silenced immediately after investigation of a novel adult mouse. When reintroduced to the now familiar adult mouse 6 hours later, a time at which studies have shown that the effects of CNO on other behaviors have worn off (Alexander et al., 2009; Whissell et al., 2016), mice exhibited control-like social discrimination (**Figure 2h**), suggesting that abGC activity may not be required for social memory consolidation.

Our findings suggest that abGC projections to the CA2 play a role in retrieval of remote memories of the mother. However, to date, no studies have investigated how abGC inputs affect CA2 network activity. Because neuronal oscillations, including SWRs and PAC, in the CA2 have been linked to social recognition of recently encountered conspecifics (Oliva et al., 2020; Zhu et al., 2022), we next investigated whether abGC inputs to the CA2 modulate these events during exposure to mothers and novel mothers, specifically.

abGCs are necessary for differential network activity in the CA2 during exposure to novel versus familiar social stimuli

To test whether abGCs affect CA2 SWRs, we recorded neuronal oscillations in the CA2 with and without inhibition of 4-6-week-old abGCs in separate groups of Nestin-cre and Nestin-cre:Gi mice (**Figure 3a**). We found that SWR production (**Figure 3c**) is increased during exposure to a social stimulus (**Figure 3e**), and that more SWRs are produced during novel mother exposure, presumably during encoding social memories, than during mother exposure, presumably during retrieval of developmental social memories (**Figure 3f**; **Supplementary Figure 4a, b**). However, inhibition of abGCs in the presence of a social stimulus altered several features of SWR production, including decreasing SWR frequency, integral, and duration (**Figure 3g**, **Supplementary Figure 3a-c**, **Supplementary Figure 4a**), but no effects on SWR peak amplitude (**Figure 3h**). Inhibition of abGCs during baseline recordings showed no change in SWR frequency, but an increase in SWR peak amplitude (**Figure 3i**). Conversely, inhibition of 10-12-week-old abGCs had no statistically significant effect on social novelty-induced increases in CA2 SWRs, nor did it affect other features of SWR production (**Figure 3k-p**, **Supplementary Figure 3d, e**, **Supplementary Figure 4b**). These findings suggest that 4-6-week-old abGCs, but not 10-12-week-old abGCs, are necessary for promoting appropriate SWR responses during social novelty exploration and developmental social memory retrieval. In an additional study, we investigated if CA2 SWRs were modulated by the presence of a nonsocial stimulus (an object) (**Supplementary Figure 5a**). Surprisingly, CA2 SWR generation was suppressed upon exposure to a nonsocial stimulus (**Supplementary Figure 5b**). Inhibiting immature abGCs had no statistically significant effects on SWR features in response to a nonsocial stimulus (**Supplementary Figure 5c-f**).

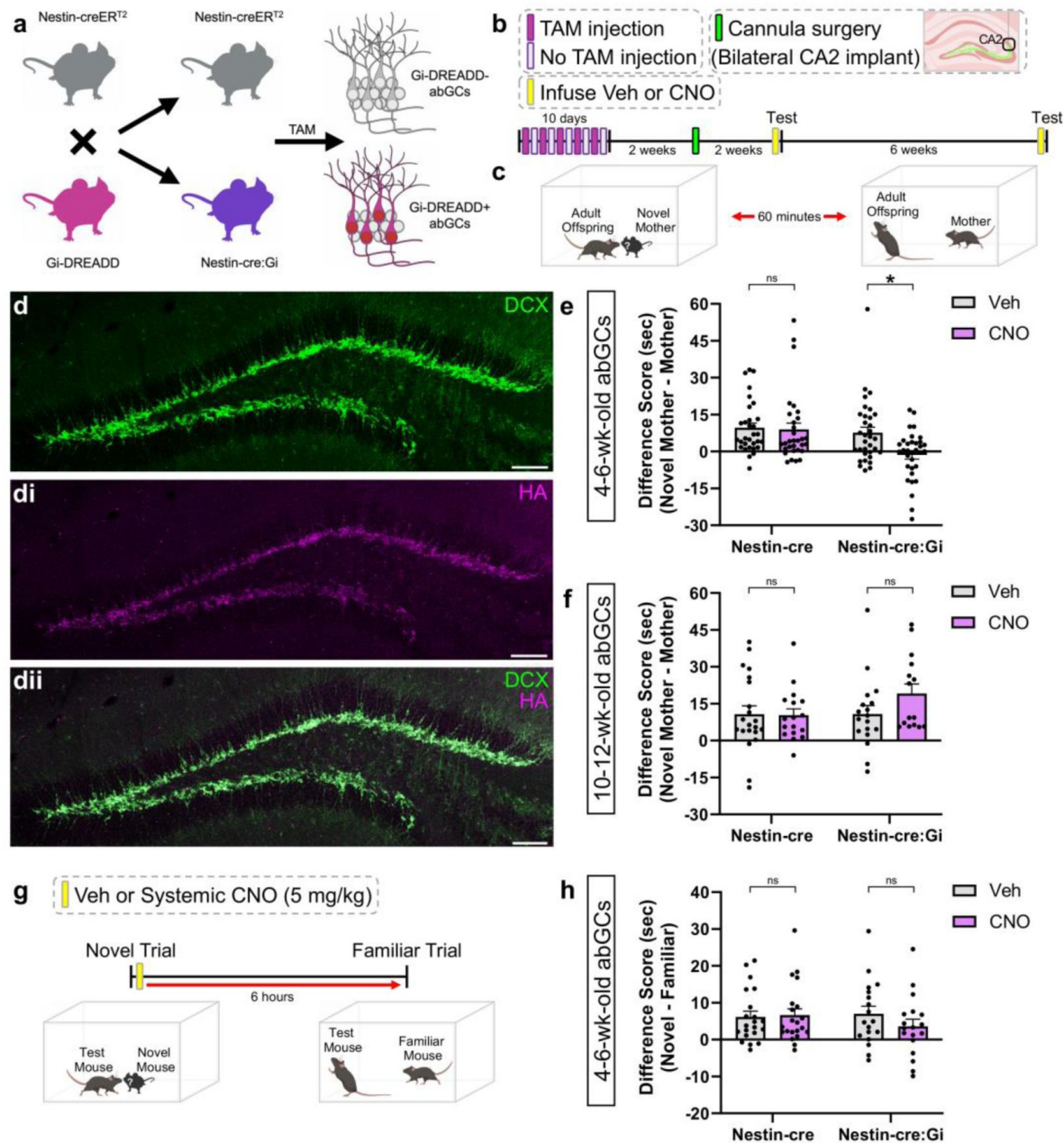


Figure 2

Inhibiting 4-6-week-old abGC projections to CA2 prevents discrimination between mothers and novel mothers.

(a) Schematic of breeding to produce Nestin-cre:Gi double transgenic offspring. (b) Timeline outlining tamoxifen administration, cannula implantation, and behavioral testing. (c) Schematic demonstrating direct social interaction testing. (d-dii) Confocal images depicting 4-6-week-old Gi-DREADD+ abGCs in the DG. (e) Inhibiting 4-6-week-old abGC projections to CA2 with CNO prevents discrimination between mother and novel mother (Nestin-cre: $n = 31$; Nestin-cre:Gi: $n = 33$; 2-way RM ANOVA: Genotype $\times$ Drug: $F_{1,62} = 4.042$, $p = 0.0487$; Šidák's test $p = 0.0057$). (f) Inhibiting 10-12-week-old abGC projections to CA2 does not influence discrimination between mother and novel mother (Nestin-cre: $n = 21$ (Veh), $n = 17$ (CNO); Nestin-cre:Gi: $n = 18$ (Veh), $n = 15$ (CNO); Mixed-effects model: Genotype $\times$ Drug: $F_{1,67} = 1.669$, $p = 0.2008$). (g) Schematic for experiment assessing abGC contributions to social memory consolidation. (h) Systemic inhibition of abGCs during memory consolidation has no significant effect on social memory (Nestin-cre: $n = 21$; Nestin-cre:Gi: $n = 18$; 2-way RM ANOVA: Genotype $\times$ Drug: $F_{1,37} = 1.238$, $p = 0.2730$). * $p < 0.05$, bars represent mean + SEM. ns = not significant. Scale bars in d = 200 μm

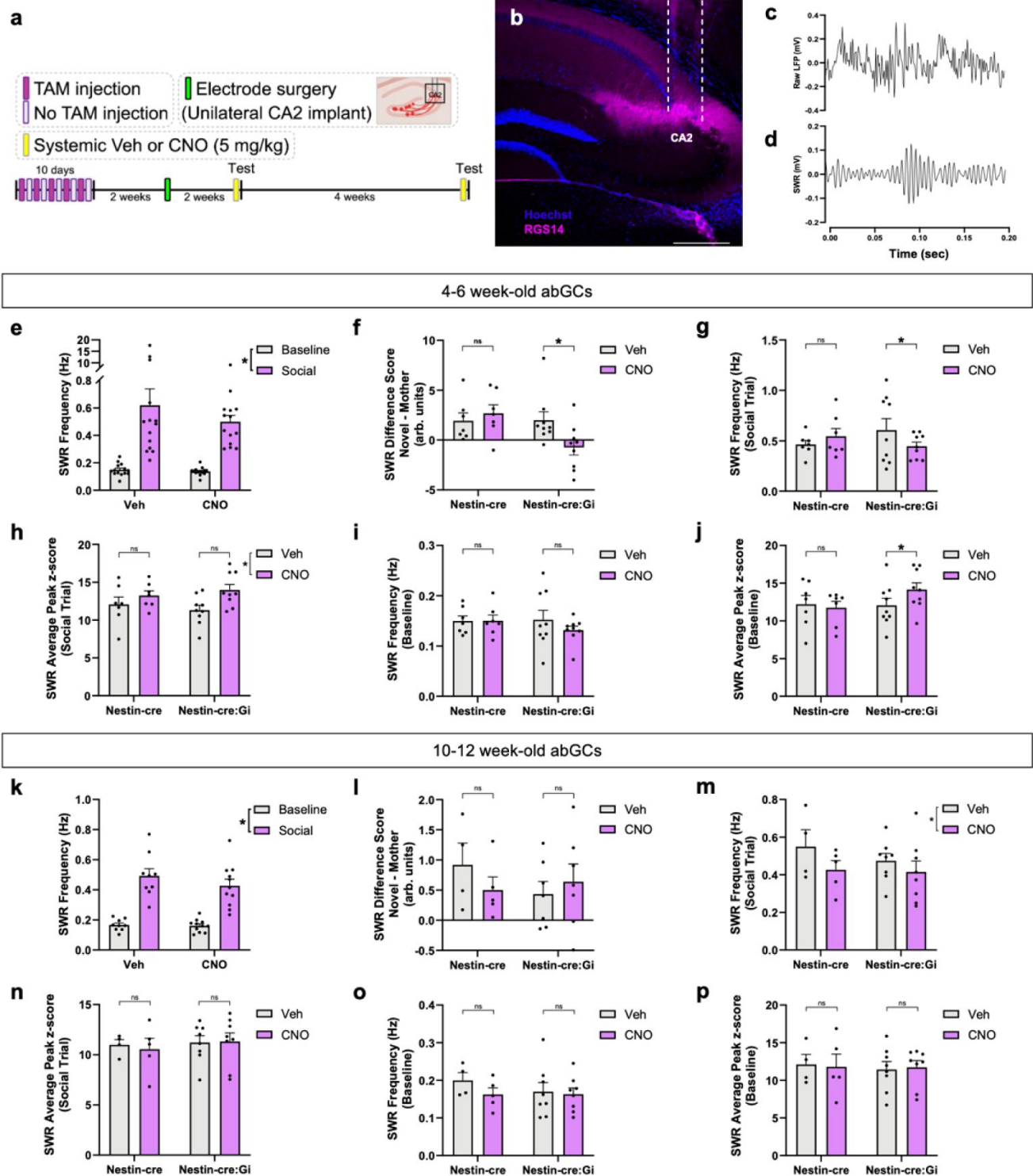


Figure 3

Adult-born neurons influence CA2 SWR changes associated with discrimination between novel mother and mother

(a) Timeline of experiment demonstrating tamoxifen injection schedule and electrode implantation in CA2. (b) Confocal image demonstrating accuracy of electrode placement. (c) Schematic of electrophysiological recordings during baseline and social trials. (d) Example SWR raw trace (top) and filtered trace (bottom). (e) CA2 SWR frequency is increased during social interactions, regardless of whether the stimulus is the novel mother or mother (Veh: $n = 14$; CNO: $n = 14$; 2-way RM ANOVA: Stimulus: $F_{1,13} = 46.40$, $p < 0.0001$). (f) Inhibiting 4-6-week-old abGCs with CNO prevents characteristic SWR production patterns present during exposure to novel mothers vs mothers (Nestin-cre: $n = 7$; Nestin-cre:Gi: $n = 9$; Mixed-effects model: Genotype x Drug: $F_{1,110} = 18.6529$, $p < 0.0001$; Tukey's test $p < 0.0001$). (g) Inhibiting 4-6-week-old abGCs significantly reduces social trial SWR production (Nestin-cre: $n = 7$; Nestin-cre:Gi: $n = 9$; Mixed-effects model: Genotype x Drug: $F_{1,228} = 16.2465$, $p < 0.0001$; Tukey's test $p = 0.0018$). (h) Inhibiting 4-6-week-old abGCs increases SWR peak amplitude during social interaction trials (Nestin-cre: $n = 7$; Nestin-cre:Gi: $n = 9$; Mixed-effects model: Genotype x Drug: $F_{1,238} = 3.69$, $p = 0.0559$) and (j) baseline (Nestin-cre: $n = 7$; Nestin-cre:Gi: $n = 9$; Mixed-effects model: Genotype x Drug: $F_{1,237} = 6.0322$, $p = 0.01477$; Tukey's test $p = 0.0133$). (i) Changes in baseline SWR production did not reach significance (Nestin-cre: $n = 7$; Nestin-cre:Gi: $n = 9$; Mixed-effects model: Genotype x Drug: $F_{1,238} = 3.5101$, $p = 0.0622$). (k) At the 10-12 wk abGC time point, CA2 SWR frequency is increased during social interactions, regardless of stimulus (Veh: $n = 9$; CNO: $n = 11$; Mixed-effects model: Stimulus: $F_{1,10} = 133.1$, $p < 0.0001$). (l) Inhibiting 10-12-week-old abGCs has no influence on characteristic SWR production patterns present during exposure to the novel mother vs mother (Nestin-cre: $n = 4$ (Veh), $n = 5$ (CNO); Nestin-cre:Gi: $n = 7$ (Veh and CNO); Mixed-effects model: Genotype x Drug: $F_{1,82} = 1.7431$, $p = 0.1904$), nor on (m) SWR production during social trials (Nestin-cre: $n = 4$ (Veh), $n = 5$ (CNO) Nestin-cre:Gi: $n = 8$ (Veh and CNO); Mixed-effects model: Genotype x Drug: $F_{1,170} = 0.4496$, $p = 0.503$), or (o) during baseline recording (Nestin-cre: $n = 4$ (Veh), $n = 5$ (CNO); Nestin-cre:Gi: $n = 8$ (Veh and CNO); Mixed-effects model: Genotype x Drug: $F_{1,170} = 2.0552$, $p = 0.15352$). (n) Inhibiting 10-12-week-old abGCs has no influence on SWR peak amplitude social interaction trials (Nestin-cre: $n = 4$ (Veh), $n = 5$ (CNO); Nestin-cre:Gi: $n = 8$ (Veh and CNO); Mixed-effects model: Genotype x Drug: $F_{1,18} = 0.9744$, $p = 0.3250$) or (p) during baseline (Nestin-cre: $n = 4$ (Veh), $n = 5$ (CNO); Nestin-cre:Gi: $n = 8$ (Veh and CNO); Mixed-effects model: Genotype x Drug: $F_{1,172} = 0.540$, $p = 0.8165$), * $p < 0.05$, bars represent mean + SEM. Scale bar in **b** = 200 μm . ns = not significant.

We next investigated abGC contributions to theta-gamma PAC (**Figure 4a**), which is increased after exposure to social stimuli (Zhu et al., 2022) and has been linked to nonsocial memory retrieval (Tronel et al., 2015). We found that exposure to the mother was associated with increased theta-mid-gamma PAC, and that this increase was prevented after inhibition of 4-6-week-old abGCs (**Figure 4b-d**). Conversely, 4-6-week-old abGCs had no influence on CA2 oscillation power across theta, low-gamma, and mid-gamma frequency bands (**Supplementary Figure 6a-f**). Inhibition of a 10-12-week-old cohort of abGCs had no statistically significant effect on PAC signatures evoked during exposure to the mother (**Figure 4e-g**), although there appears to be an overall shift in gamma phase within theta cycles, especially during the mother exposure condition (**Figure 4g**). This suggests that abGC influence on CA2 network activity is limited to a specific abGC age range. Taken together with our SWR results, these findings suggest that 4-6-week-old abGCs support CA2 network oscillations present during both encoding and retrieval of social memories.

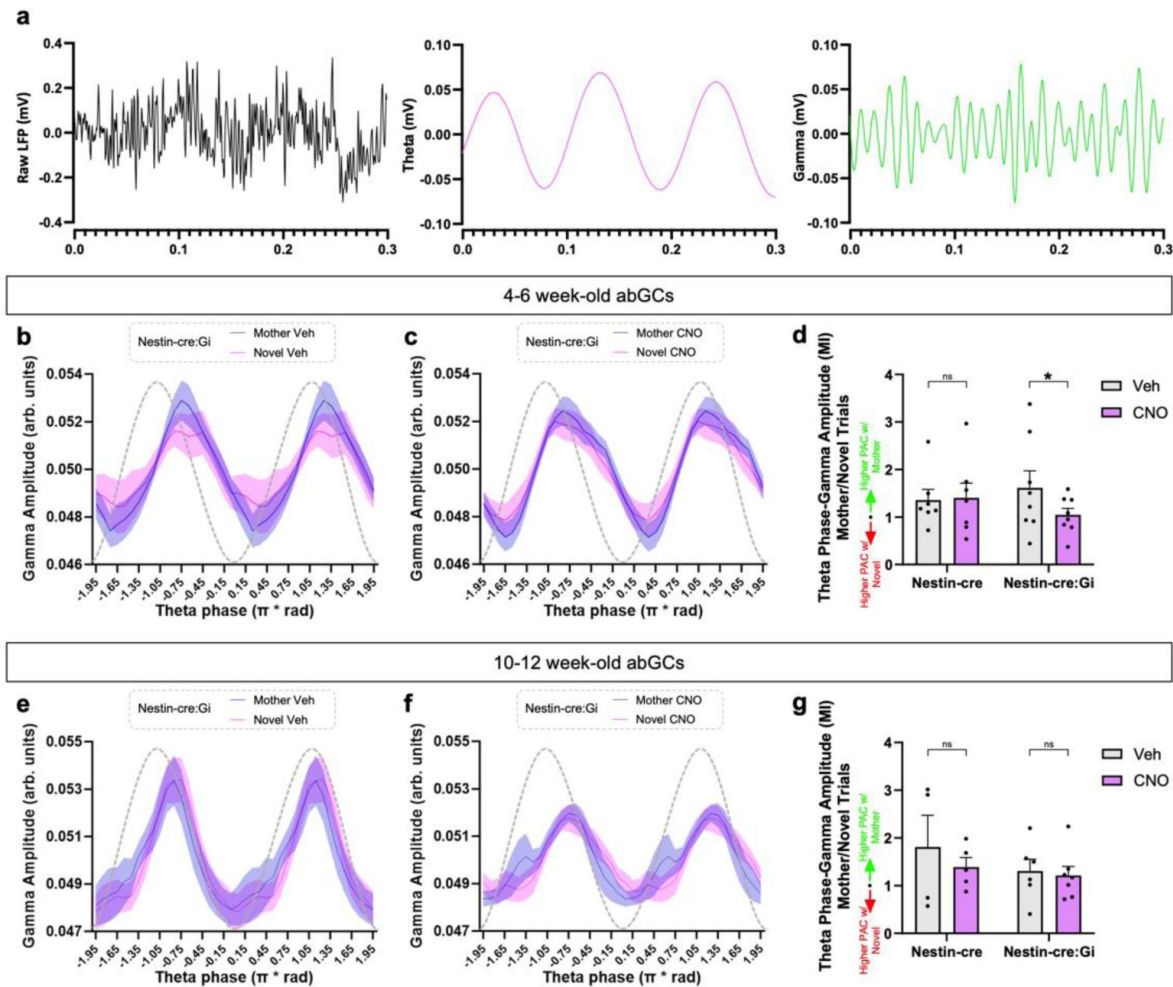


Figure 4

Adult-born neurons influence CA2 PAC changes associated with retrieval of developmental memories of the mother

(a) Example trace of raw LFP (black), 4-12 Hz theta oscillation (magenta), and 50-100 Hz gamma oscillation (green). (b,c) Graphs demonstrating mid gamma amplitude modulation across theta phases in Nestin-cre:Gi mice. (d) Inhibiting 4-6-week-old abGCs abolishes the increase in theta-mid gamma PAC present during exposure to the mother (Nestin-cre: $n = 7$; Nestin-cre:Gi: $n = 8$; Mixed-effects model: Genotype $\times$ Drug: $F_{1,103} = 4.6729$, $p = 0.03296$; Šídák's test $p = 0.0219$). (e,f) Graphs demonstrating mid gamma amplitude modulation across theta phases in Nestin-cre:Gi mice. (g) Inhibiting 10-12-week-old abGCs has no influence on theta-mid gamma PAC during exposure to the mother (Nestin-cre: $n = 4$ (Veh), $n = 5$ (CNO); Nestin-cre:Gi: $n = 8$ (Veh), $n = 7$ (CNO); Mixed-effects model: Genotype $\times$ Drug: $F_{1,18} = 0.2616$, $p = 0.6153$).

Discussion

Hippocampal CA2 plays an important role in supporting social recognition (Hitti and Siegelbaum, 2014 [↗](#); Smith et al., 2016 [↗](#); Laham et al., 2021 [↗](#); Diethorn and Gould, 2023 [↗](#)), yet the contributions of the abGC projection to this region have remained uninvestigated. Here we show that ablation of abGCs impairs the retrieval of developmental memories of the mother. Recovery of abGCs and their projections to CA2 restored the ability to discriminate between novel mothers and caregiving mothers, revealing that abGCs support developmental social memory retrieval and not remote memory storage. Implementing a double transgenic mouse model, we found that chemogenetic inhibition targeted at the 4-6-week-old abGC projections to the CA2 prevents the ability to discriminate between mothers and novel mothers. This effect was not detected when inhibiting 10-12-week-old abGC projections, revealing that abGCs possess a time-sensitive ability to support retrieval of developmental mother memories.

It is important to note that we cannot rule out the possibility that projections from abGCs to the CA3 were also affected in double transgenic mice with CNO infusion, due to the potential diffusion of the drug away from the CA2 cannula site. Along these lines, studies have shown that the ventral CA3 is also involved in social recognition (Chiang et al., 2018 [↗](#)) and that areas CA2 and CA3 may act together to coordinate replay (Stöber et al., 2020 [↗](#)). Thus, the abGC-CA2 circuit may work in concert with the abGC-CA3 circuit, followed by crosstalk between the two target areas to mediate social recognition.

We also investigated if abGCs support developmental mother memory retrieval via modulation of CA2 network activity and found that exposure to novel mothers produced a larger increase in CA2 SWR frequency than exposure to mothers. We also found the converse with CA2 theta-mid-gamma PAC, which was higher during exposure to mothers than to novel mothers. Both effects were significantly diminished after inhibiting 4-6-week-old abGCs. Thus, inhibition of immature abGCs alters SWR generation and theta-mid-gamma PAC in the CA2 region in a way that increases network activity similarity across novel and familiar social interactions. These changes likely affect broader hippocampal circuitry as inhibiting CA2 activity alters both SWRs and gamma oscillations in CA1 (Alexander et al., 2018 [↗](#); Brown et al., 2020 [↗](#)). abGC inhibition of network orthogonalization may contribute to the inability to socially discriminate.

Our data suggest that abGCs contribute to the diversification of neuronal oscillatory states in the CA2 in the service of both recognizing social novelty and retrieving developmental social memories. The mechanisms underlying these effects remain unknown. Although we did not record single units along with neuronal oscillations, previous studies suggest that CA2 SWRs correspond with the activation of neuronal ensembles that are associated with previous social experience, at least during sleep (Oliva et al., 2020 [↗](#)). Thus, by promoting the generation of SWRs during social, especially novel, experience, abGCs may provide a network conducive to replay. The ability of abGCs to facilitate the generation of SWRs and PAC may be related to their preferential connections to PV+ interneurons, which have been causally linked to SWRs, as well as to theta, and gamma oscillations (Schlingloff et al., 2014 [↗](#); Stark et al., 2014 [↗](#); Amilhon et al., 2015 [↗](#); Gan et al., 2017 [↗](#); Antonoudiou et al., 2020 [↗](#); Park et al., 2020 [↗](#); Kriener et al., 2022 [↗](#)). A time-limited role for abGCs in nonsocial memory has been previously reported (Gu et al., 2012 [↗](#); Song et al., 2012 [↗](#); Swan et al., 2014 [↗](#); Kropff et al., 2015 [↗](#)). Our results raise questions about how these cells change as they mature, as well as what the function of abGC projections to the CA2 might be after their involvement in remote mother memory retrieval has ended. Previous studies have shown that abGCs are highly excitable and exhibit enhanced synaptic plasticity 4-6 weeks after their generation, and that these properties appear to be diminished by 8 weeks post-mitosis (Song et al., 2012 [↗](#); Toni and Schinder, 2015 [↗](#)). As abGCs mature and become more inhibited, they may function to provide input to the CA2 only when social stimuli are especially salient, such as when they are threatening or associated with a potent reward. It may also be relevant to determine whether abGC inputs to the CA2 affect synaptic input from the supramammillary nucleus of the

hypothalamus, which is known to respond to social novelty (Chen et al., 2020) and drive novelty-induced increases in adult neurogenesis (Li et al., 2022 [↗](#)). It is possible that abGCs influence this circuit by increasing excitatory tone in the same target cells, which in turn alters neuronal oscillations, although this hypothesis remains untested.

Materials and Methods

Animals

All animal procedures were approved by the Princeton University Institutional Animal Care and Use Committee and were in accordance with the National Research Council Guide for the Care and Use of Laboratory Animals. C57 male and female mice were obtained from Jackson laboratories and were used for immunolabeling. Transgenic mice expressing herpes simplex virus thymidine kinase (TK) under the GFAP promoter were bred in the Princeton Neuroscience Institute animal colony with founders provided by Dr. Heather Cameron at the National Institute of Mental Health for adult neurogenesis ablation studies. GFAP-TK mice were generated by crossbreeding CD1 male mice with heterozygous GFAP-TK female mice (Snyder et al., 2011 [↗](#)). Nestin-CreERT2 and R26LSL-hM4Di single transgenic mice were obtained from Jackson labs and bred in the Princeton Neuroscience Institute as single or double transgenic mice for cannula implantation and electrophysiology experiments. PV-cre mice were obtained from Jackson labs and bred in the Princeton Neuroscience Institute to investigate abGC projections to CA2 PV+ interneurons. All studies used mixed-sex groups. Transgenic mice were genotyped by Transnetyx from ear punches at P15 using real-time PCR, separated from their dams at P21 and housed in same sex groups. GFAP-TK and CD1 mice were treated with VGCV starting around P60. Nestin-cre and nestin-cre:Gi transgenic mice were subjected to craniotomy for bilateral cannula implantation, or unilateral electrode implantation starting around P60. GFAP-TK mice were group housed by genotype as well as sex to avoid any potential behavioral differences after ablation of abGCs (Tsuda et al., 2023 [↗](#)), while single and double transgenic animals were group housed with mixed genotypes. Animals were housed in Optimice cages on a reverse 12/12 h light/dark cycle.

Behavioral testing

The direct social interaction test was used (Laham et al., 2021 [↗](#)) to determine whether adult offspring were able to discriminate between their mother and a novel mother. Novel mothers were female mice of the same age and reproductive experience as the mothers, which the adult offspring had never encountered. To avoid the confound of altered social preference during times of sexual receptivity, the estrous cycle was tracked in the mother and novel mother, and behavioral testing only occurred when stimulus mice were in diestrus (Cora et al., 2015 [↗](#)). Adult GFAP-TK, Nestin-cre, and Nestin-cre:Gi offspring underwent a social interaction test in which they directly interacted with the mother, or a novel mother for 5 minutes. After a 1-hour delay spent in the home cage with cagemates, mice were introduced to the stimulus mouse not previously encountered for 5 minutes. The order of stimulus mouse exposure was counterbalanced in all tests. All behavioral tests were recorded, and a trained observer scored behavior testing under blind conditions. Testing apparatuses were cleaned with 70% ethanol after each trial. Bouts of investigation were characterized as direct sniffing of the stimulus mouse's anogenital region, body, and head.

An additional experiment explored abGC contributions to social memory consolidation. Nestin-cre and Nestin-cre:Gi mice were placed in a testing apparatus with a novel adult mouse and were permitted to investigate for 5 minutes. Immediately after the interaction, mice received a systemic injection of Veh or 5 mg/kg CNO and were returned to their cages in the vivarium for 6 hours. At the conclusion of the 6-hour intertrial interval, mice were returned to the testing apparatus housing the stimulus mouse they had previously encountered and allowed to investigate for an additional 5 minutes. Previous studies have shown that 6 hours after CNO treatment, other types

of behavioral changes return to baseline and baseline neuronal activity is largely restored (Alexander et al., 2009 [DOI](#); Whissell et al., 2016 [DOI](#)). For each experiment that involved a social interaction measure, the time spent moving was recorded. Total locomotion was divided by trial duration to create a % locomotion measure.

Surgical procedures

Mice were deeply anesthetized with isoflurane (2-3%) and placed in a stereotaxic apparatus (Kopf) under a temperature-controlled thermal blanket for all surgeries. The head was leveled using bregma, lambda, and medial-lateral reference points before virus injection or implantation of cannula or electrode. PV-cre mice received bilateral injections of a cre-dependent mCherry AAV into CA2 (AP: -1.82, ML: +/-2.15, DV: -1.67) through a WPI nanofil 33 gauge beveled needle. Nestin-cre and Nestin-cre:Gi mice were bilaterally implanted with cannulae (Plastics One, Cat# C315GS-5/SP) targeting CA2. Dummy cannulae (Plastics One, Cat# C315DCS5/SPC) were tightened inside guides prior to implantation. After lowering to the desired target region, cannulae were secured in place using metabond followed by dental cement (Bosworth Trim). For CA2 recordings, Nestin-cre and Nestin-cre:Gi mice were implanted unilaterally with a custom-made 4-wire electrode array (Microprobes) into the right hemisphere targeting CA2. 4 bone screws were implanted into the skull with 1 screw positioned on the contralateral hemisphere serving as ground. The ground wire was wrapped around the ground screw and covered with metallic paint to ensure maximum contact. Electrode implants were kept in place using metabond followed by dental cement (Bosworth Trim). Two weeks after surgery, cannula mice were infused with either vehicle or CNO (see Drug treatments) before being tested on behavioral tasks (see Behavioral testing); electrode mice received systemic injections of vehicle or CNO 30 minutes prior to electrophysiological recordings (see Electrophysiology recordings).

Drug treatments

Administration of the antiviral drug valganciclovir (VGCV) ablates adult neurogenesis in GFAP-TK animals. CD1 and GFAP-TK mice underwent behavioral testing at 3 time points. Animals first underwent testing prior to VGCV administration (VGCV-trial). After the first round of behavioral testing, VGCV was added to powdered rodent chow (227 mg of VGCV per kg chow) for 6 weeks. After 6 weeks of consuming VGCV chow, animals underwent behavioral testing for a second time (VGCV+ trial). At the conclusion of the VGCV+ trial, VGCV chow was removed from all cages and replaced with standard chow. After 6 weeks of standard chow consumption, animals underwent a third round of behavioral testing (VGCV-recovery). Mice were perfused shortly after the third round of behavioral testing to assess the extent of adult neurogenesis recovery.

Cannula-implanted Nestin-cre and Nestin-cre:Gi mice underwent social discrimination testing (with novel mother and mother stimulus mice) twice, once after vehicle cannula infusion and once after CNO cannula infusion. The order of drug administration (vehicle or CNO) was counterbalanced across groups. Electrode-implanted mice underwent direct social interaction testing with systemic vehicle or CNO administration. Mice were given a 48-hour minimum rest period between vehicle and CNO tests. 30 minutes prior to the first stimulus mouse exposure, test mice received vehicle or CNO cannula infusions. For cannula infusions, 200 nl of vehicle or CNO (2 µg/µl of CNO dissolved in DMSO suspended in saline) (Chang and Gean, 2019 [DOI](#)) was infused per hemisphere over 1 minute into CA2 using a syringe pump (Harvard apparatus) mounted with a 1 µl syringe (Hamilton). The internal cannulae remained in place for 1 additional minute after the infusion was completed to allow for diffusion of the drug.

Electrophysiology recordings

Local field potentials (LFPs) were recorded using a wireless head stage (TBSI, Harvard Biosciences). To habituate mice to the weight of the recording head stage, mice were connected to a custom head stage with equivalent weight while in the home cage for 10 minutes a day for 5

consecutive days. The behavioral testing paradigm consisted of a 3-minute baseline followed by a 5-minute social interaction period. Animals underwent this recording paradigm for both mother and novel mother trials. Vehicle or 5mg/kg CNO was administered via IP injection 30 minutes before testing. In an additional experiment, Nestin-cre and Nestin-cre:Gi mice with CA2 electrodes were recorded during a 1-minute baseline followed by a 2-minute nonsocial object exposure trial. The nonsocial stimulus was a plastic toy animal of similar size to the mouse. Both baseline and nonsocial stimulus trials took place in an apparatus identical to that of the social stimulus experiments. The nonsocial stimulus trials were conducted after the conclusion of all social testing at the 4-6-week-old abGC time point. LFPs were recorded continuously throughout baseline and stimulus trials. The data were transmitted to a wireless receiver (Triangle Biosystems) and recorded using NeuroWare software (Triangle Biosystems).

Sharp wave-ripple analysis

All recordings were processed using Neuroexplorer software (Nex Technologies) and custom Python scripts (Laham and Zahn, 2023 [DOI](#)). For SWR analyses, continuous LFP data were notched at 60 Hz and band-pass filtered between 140 and 220 Hz. Signal underwent Hilbert transform before being z-scored. Using a custom Python script, SWRs were considered as events exceeding 3 standard deviations for a minimum of 15 ms. SWRs occurring within 15 ms of one another were merged into a single SWR event. SWR frequency, peak amplitude, integral, duration, and raw number were normalized to the respective baseline trial. SWR integral, which provides information about the average total power of SWR events and has been shown to influence SWR propagation (De Filippo and Schmitz, 2023 [DOI](#)) was calculated by summing z-scored amplitude values from beginning to the end of each SWR envelope, and then averaging these values across all summed envelopes. SWR measures were determined for baseline, novel mother, and mother trials for the entirety of each testing period.

Phase-amplitude coupling analysis

Theta (4-12 Hz), low-gamma (25-50 Hz), and mid-gamma (50-100 Hz) oscillations were extracted from the raw LFP signal obtained during the entire baseline, novel mother exposure, and mother exposure recordings using a custom Python script. Theta and mid-gamma recordings underwent a Hilbert transform, and theta phase and mid-gamma analytic signal were stored. PAC was quantified using a Modulation Index (Tort et al., 2010 [DOI](#)).

Histology

Mice were deeply anesthetized with Euthasol (Virbac) and were transcardially perfused with cold 4% paraformaldehyde (PFA). Extracted brains were postfixed for 48 h in 4% PFA at 4°C followed by an additional 48 h in 30% sucrose at 4°C for cryoprotection before being frozen in cryostat embedding medium at -80°C. Hippocampal coronal sections (40 µm) were collected using a cryostat (Leica). Sections were blocked for 1.5 hours at room temperature in a PBS solution that contained 0.3% Triton X-100 and 3% normal donkey serum. Sections were then incubated overnight while shaking at 4°C in the blocking solution that contained combinations of the following primary antibodies: mouse anti-3 microtubule-binding domain tau protein (3R-Tau, 1:500, Millipore, Cat# 05-803), rabbit anti-Purkinje cell protein 4 (PCP4, 1:500, Sigma-Aldrich, Cat# HPA005792), rabbit anti-zinc transporter 3 (ZnT3, 1:500, Alomone labs, Cat# AZT-013). For 3R-Tau immunohistochemistry, sections were subjected to an antigen retrieval protocol that involved incubation in sodium citrate and citric acid buffer for 30 minutes at 80°C prior to blocking solution incubation. Washed sections were then incubated for 1.5 hours at room temperature in secondary antibody solutions that contained combinations of the following secondaries: donkey anti-rat Alexa Fluor 568 (1:500, Abcam), donkey anti-mouse Alexa Fluor 568 or 647 (1:500, Invitrogen), or donkey anti-rabbit Alexa Fluor 488 (1:500, Invitrogen). Washed sections were then counterstained with Hoechst 33342 for 10 minutes (1:5,000 in PBS, Molecular Probes), mounted onto slides, and

coverslipped with Vectashield (Vector labs). Slides were coded until completion of the data analysis. Sections through the dorsal hippocampus from cannula and electrophysiology studies were stained for Hoechst 33342 to verify accurate cannula and electrode placement.

Optical intensity measurements

Z-stack images of the CA2 and corpus callosum were taken using a 40x objective with a 0.5 μ m step size through the entire 40 μ m stack on a Leica SP8 confocal with LAS X software (version 35.6). The CA2 was defined by PCP4 labeling. Collected z-stack images were analyzed for optical intensity in Fiji (NIH).

For high magnification (40X objective with 3X zoom) intensity analyses of 3R-Tau labeled puncta in close proximity to dendrites of mCherry+ PV cells and dendrites of PCP4+ pyramidal cells, mCherry+ and PCP4+ labeled cell bodies were first identified in the pyramidal cell layer. Then, proximal dendrites were identified emanating from selected labeled cells and the ROI was traced around the dendrite. The dendrite was defined from the initial tapering off from the cell body and extending for 7 μ m in length, which was the maximum that could be traced to maintain certainty that the entirety of the selected dendrite belonged to a specific cell body. All dendrites selected for analysis were located either in the pyramidal cell layer or the stratum lucidum of CA2. For each brain, 10 labeled cells of each cell type (PV+ cells or PCP4+ pyramidal cells) were selected, and from these, 1 dendrite per cell was randomly selected for analysis per brain. The intensity of 3R-Tau labeling on dendrites was determined for each cell type using Fiji as follows: A background subtraction using a rolling ball radius (50 pixels) was applied to the image stacks. A region of interest (ROI) was drawn, and the mean gray value was collected throughout the image stack. Averages were obtained for each brain and statistical comparisons were made between intensity on PV-mCherry+ proximal dendrites and intensity on PCP4+ proximal dendrites.

For overall intensity values of 3R-Tau to assess abGC innervation of CA2, the ROI included the pyramidal cell layer and stratum lucidum. The mean gray value of the ROI was calculated for each z-slice and the maximum mean gray value for each z-stack was taken from 3 sections per brain. The maximum of the CA2 ROI was divided by the maximum of the corpus callosum ROI for each section. Averages were obtained for each brain and statistical comparisons were made between lower magnification 3R-Tau intensity across different GFAP-TK groups.

Cell density and percentage measurements

The number of 3R-Tau+ cells was counted in the dorsal dentate gyrus of the hippocampus on 4 neuroanatomically matched sections using an Olympus BX-60 microscope with a 100x oil objective. The area measurements were collected using Stereo Investigator software (MBF). The density of 3R-Tau was determined by dividing the total number of positively labeled cells by the volume of the subregion (ROI area multiplied by 40 μ m section thickness).

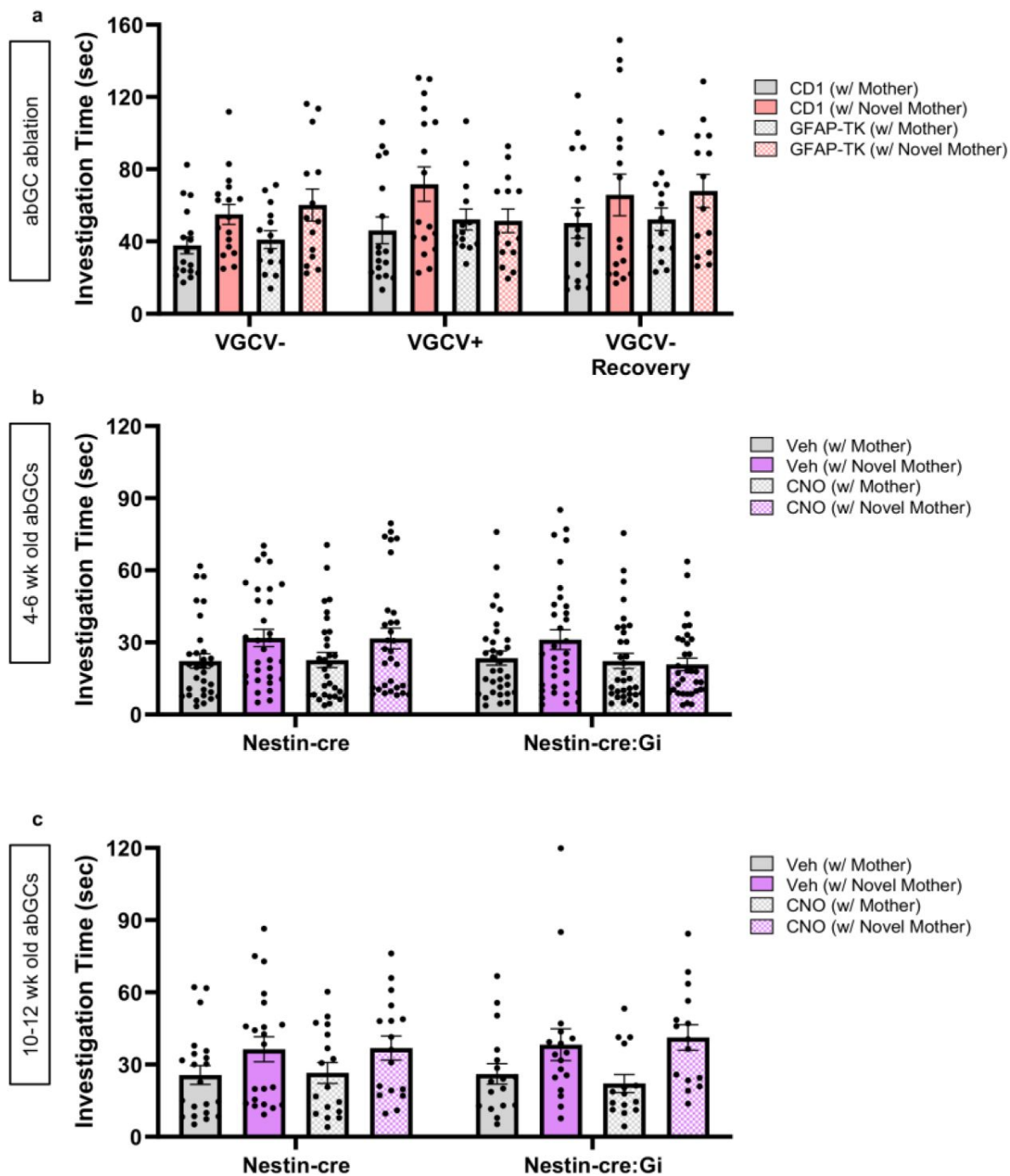
Statistical analyses

Statistical analyses are presented in the figure legends. For histological analyses, data sets were analyzed using an unpaired 2-tailed Student's t-test or Mann-Whitney U tests. For behavioral analyses involving 2 group comparisons, data sets were analyzed using either unpaired two-tailed Student's t-tests or a repeated measures 2-way ANOVA, as appropriate. For behavioral analyses involving virus manipulations, data sets were analyzed using either a 2-way ANOVA, a mixed effects model, or a repeated measures 3-way ANOVA. Bonferroni post hoc comparisons were used to follow up any significant main effects or interactions of the ANOVAs. Pearson's correlation coefficient test was used to analyze the association between theta power and social investigation times. Because electrophysiological measurements were taken from multiple electrodes within each mouse, these data were analyzed with linear mixed-effects ANOVAs using the lme4 package (Bates et al., 2015 [DOI](#)). The level of the measurement was explained by drug, virus, genotype, the 2-

way interaction, the 3-way interaction, and a random effect of mouse. Tukey post hoc comparisons were used to follow up any significant main effects or interactions using the emmeans package (Lenth, 2022 [link](#)). All data sets are expressed as the mean $\pm$ SEM on the graphs and statistical significance was set at $p < 0.05$ with 95% confidence. GraphPad Prism 9.2.0 (GraphPad Software), Excel (Microsoft), or R studio were used for statistical analyses. All graphs were prepared using GraphPad Prism 9.2.0 (GraphPad Software).

Acknowledgements

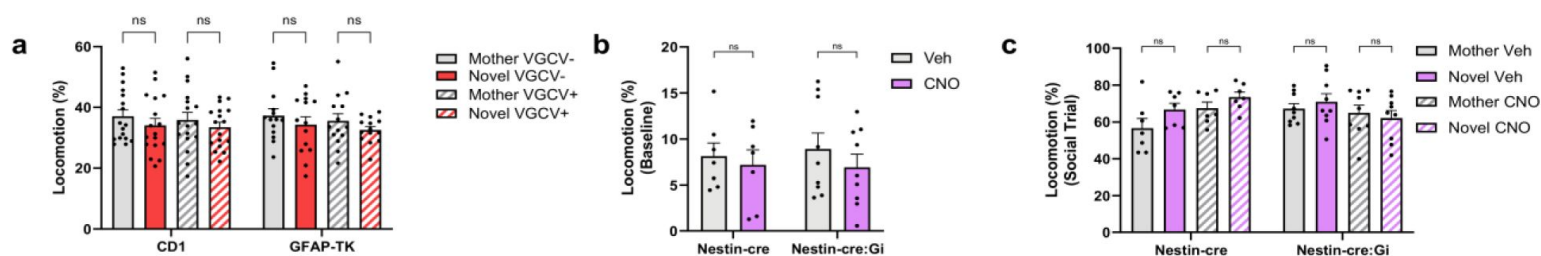
This work was supported by the National Institutes of Health, NIMH 1R01 MH118631-01 to EG. The authors thank Samantha H. Wang for performing surgeries on PV-cre mice and for help with editing the text. The authors acknowledge Biorender for assistance with the figure schematics.



Supplementary Figure 1

Total investigation times during social interaction.

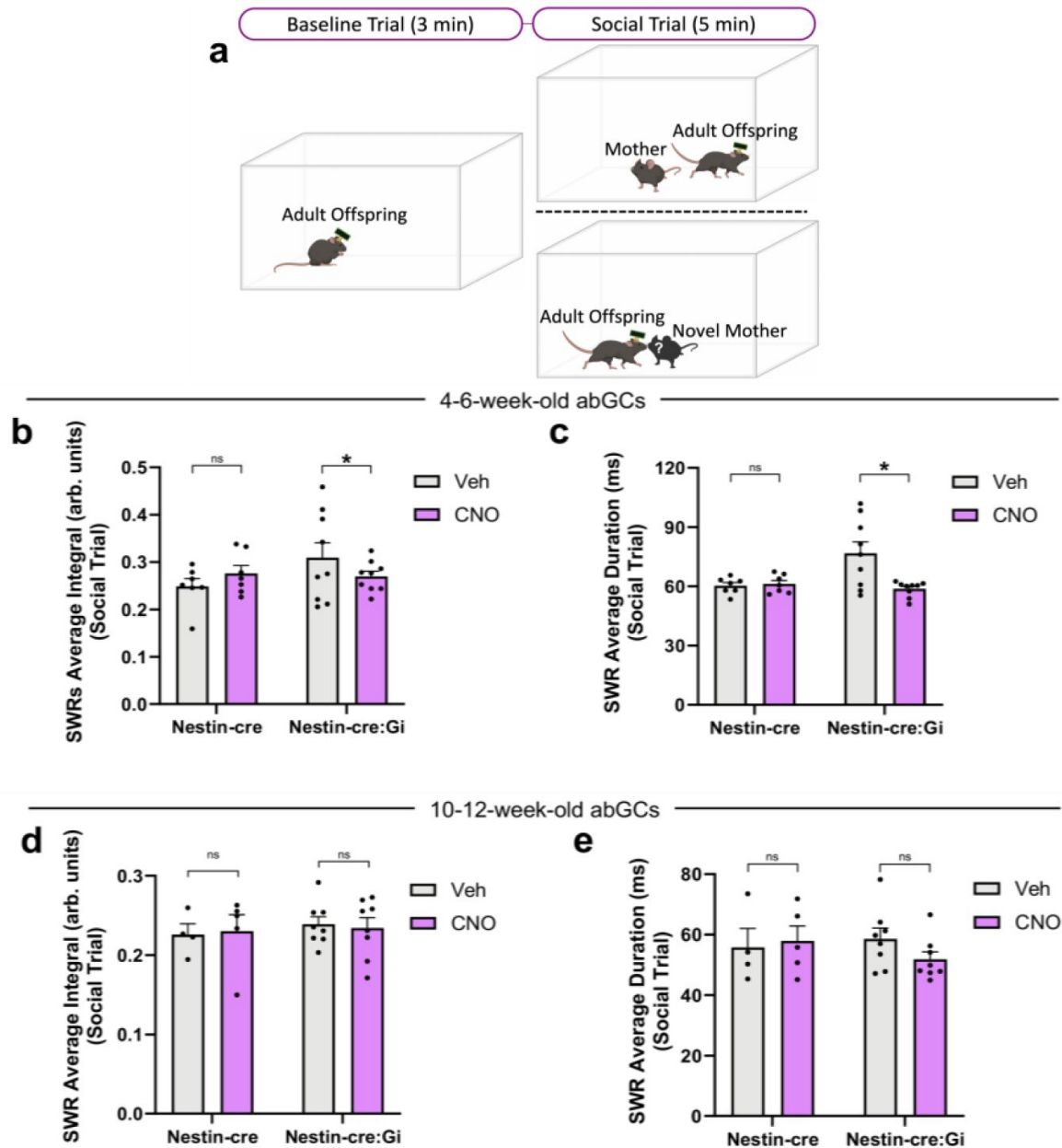
a) Total investigation times of the mother and a novel mother were assessed for CD1 and GFAP-TK mice before (VGCV-), during (VGCV+), and after VGCV administration (VGCV-Recovery) (CD1: $n = 17$; TK: $n = 14$; RM ANOVA: Genotype $\times$ Drug: $F_{6,116} = 1.672$, $p = 0.1339$; Drug: $F_{1,773,102.9} = 5.515$, $p = 0.0072$). b) Total investigation times of the mother and a novel mother were quantified for Nestin-cre mice and Nestin-cre:Gi mice with a 4-6-week-old population of inhibitory DREADD+ abGCs (Nestin-cre: $n = 31$; Nestin-cre:Gi: $n = 33$; Mixed-effects ANOVA: Genotype $\times$ Drug $\times$ Stimulus: $F_{1,56} = 4.290$, $p = 0.0430$; Tukey's test: Nestin-cre Veh Mother vs Novel $p = 0.0004$, Nestin-cre CNO Mother vs Novel $p = 0.0274$, Nestin-cre:Gi Veh Mother vs Novel $p = 0.0326$, Nestin-cre:Gi CNO Mother vs Novel $p = 0.9904$). c) Total investigation times of the mother and a novel mother were quantified for Nestin-cre mice and Nestin-cre:Gi mice with a 10-12-week-old population of inhibitory DREADD+ abGCs (Nestin-cre: $n = 21$; Nestin-cre:Gi: $n = 17$; Mixed-effects ANOVA: Genotype $\times$ Drug $\times$ Stimulus: $F_{1,12} = 1.204$, $p = 0.2941$; Drug: $F_{1,40} = 60.46$, $p < 0.0001$).



Supplementary Figure 2

VGCV or CNO administration does not alter locomotion.

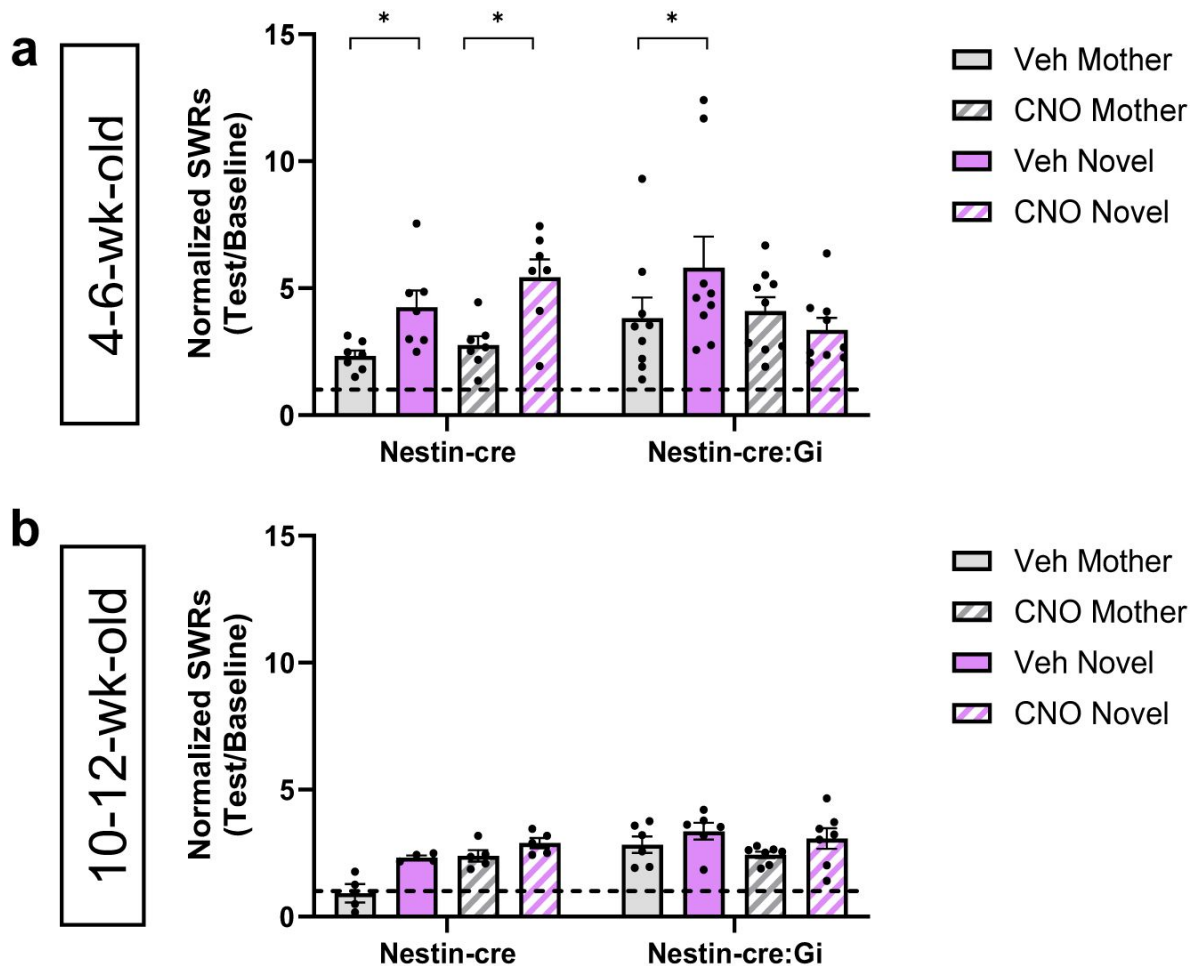
(a) GFAP-TK mice display normal locomotion during social interaction after 6 weeks of VGCV administration (CD1: $n = 17$; TK: $n = 14$; 3-way ANOVA: Genotype $\times$ Drug $\times$ Stimulus: $F_{1,29} = 0.01998$, $p = 0.8886$). CNO administration has no effect on locomotion during (b) baseline (Nestin-cre: $n = 7$; Nestin-cre:Gi: $n = 9$; 2-way RM ANOVA: Genotype $\times$ Drug: $F_{1,14} = 0.02291$, $p = 0.8818$) or (c) social stimulus trials (Nestin-cre: $n = 7$; Nestin-cre:Gi: $n = 9$; 3-way ANOVA: Genotype $\times$ Drug $\times$ Stimulus: $F_{1,14} = 0.06126$, $p = 0.8081$) in Nestin-cre and Nestin-cre:Gi mice. A significant 2-way interaction was observed during social trials, but all post hoc tests failed to reach significance (2-way ANOVA: Genotype $\times$ Drug: $F_{1,14} = 7.180$, $p = 0.0180$). Bars represent mean $\pm$ SEM. ns = not significant.



Supplementary Figure 3

4-6-week-old abGCs influence multiple features of CA2 SWRs.

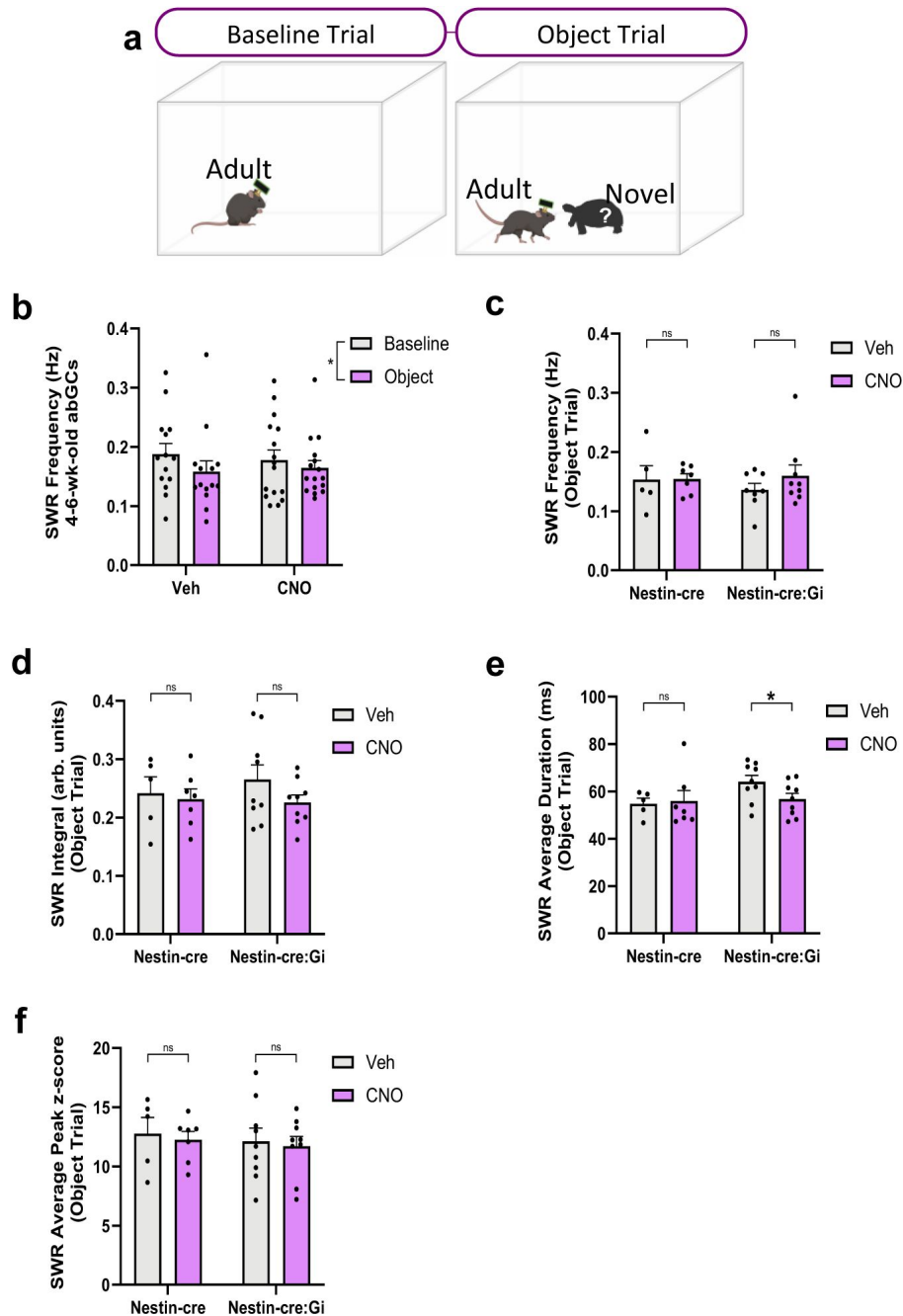
(a) Schematic of electrophysiological recordings during baseline and social trials during the 4-6-week-old abGC timepoint. (b) Silencing 4-6-week-old abGCs reduces average CA2 SWR integral (Nestin-cre: $n = 7$; Nestin-cre:Gi: $n = 9$; Mixed-effects model: Genotype $\times$ Drug: $F_{1,236} = 12.83$, $p = 0.0004017$; Tukey's test $p = 0.0099$). (c) Silencing 4-6-week-old abGCs reduces average CA2 SWR duration (Nestin-cre: $n = 7$; Nestin-cre:Gi: $n = 9$; Mixed-effects model: Genotype $\times$ Drug: $F_{1,223} = 24.077$, $p < 0.0001$; Tukey's test $p < 0.0001$). (d) Silencing 10-12-week-old abGCs has no effect on average CA2 SWR integral but a significant interaction is observed (Nestin-cre: $n = 4$ (Veh), $n = 5$ (CNO); Nestin-cre:Gi: $n = 8$ (Veh and CNO); Mixed-effects model: Genotype $\times$ Drug: $F_{1,172} = 4.0061$, $p = 0.0469$; Tukey's test $p = 0.7457$). (e) Silencing 10-12-week-old abGCs does not reduce average CA2 SWR duration but a significant interaction is observed (Nestin-cre: $n = 4$ (Veh), $n = 5$ (CNO); Nestin-cre:Gi: $n = 8$ (Veh and CNO); Mixed-effects model: Genotype $\times$ Drug: $F_{1,172} = 8.5609$, $p = 0.003898$; Tukey's test $p = 0.1439$). $*p < 0.05$, bars represent mean $\pm$ SEM. Ns = not significant.



Supplementary Figure 4

Normalized SWR generation.

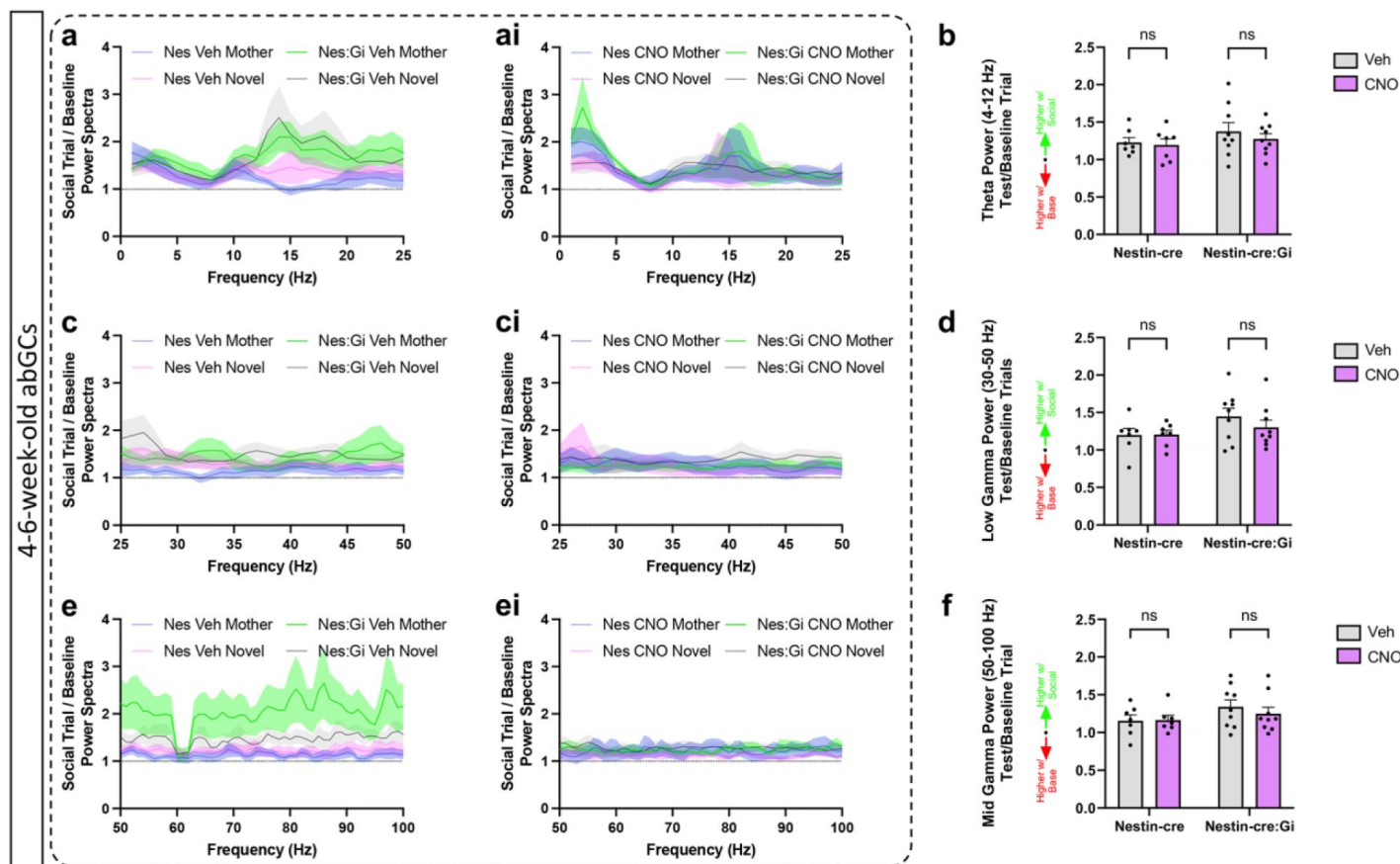
(a) Silencing 4-6-week-old abGCs alters CA2 SWR generation (Nestin-cre: $n = 7$; Nestin-cre:Gi: $n = 9$; Linear mixed-effects model: Genotype $\times$ Drug $\times$ Stimulus: $F_{1,234} = 13.3621$, $p = 0.000317$; Tukey's test: Nestin-cre Veh Mother vs Novel $p = 0.0042$, Nestin-cre CNO Mother vs Novel $p < 0.0001$, Nestin-cre:Gi Veh Mother vs Novel $p = 0.0003$, Nestin-cre:Gi CNO Mother vs Novel $p = 0.7190$). (b) Silencing 10-12-week-old abGCs has no influence on CA2 SWR generation (Nestin-cre: $n = 5$; Nestin-cre:Gi: $n = 7$; Linear mixed-effects model: Genotype $\times$ Drug $\times$ Stimulus: $F_{1,161.844} = 2.1093$, $p = 0.14834$; Stimulus: $F_{1,161.844} = 33.0013$, $p < 0.0001$). Bars represent mean $\pm$ SEM.



Supplementary Figure 5

Nonsocial stimuli suppress CA2 SWR frequency.

(a) Schematic of electrophysiological recordings during baseline and exposure to nonsocial object (plastic toy animal) trials during the 4-6-week-old abGC timepoint. (b) Nonsocial stimuli slightly inhibit CA2 SWR frequency (Veh: $n = 14$; CNO: $n = 16$; 2-way RM ANOVA: Stimulus: $F_{1,28} = 7.760$, $p = 0.0095$). (c) Silencing 4-6-week-old abGCs has no influence on CA2 SWR frequency during object exposure (Nestin-cre: $n = 5$ (Veh), $n = 7$ (CNO); Nestin-cre:Gi: $n = 8$ Veh), $n = 9$ (CNO); Mixed-effects model: Genotype x Drug: $F_{1,95} = 0.7761$, $p = 0.3806$), (d) average integral (Nestin-cre: $n = 5$, $n = 7$ (CNO); Nestin-cre:Gi: $n = 9$ (Veh and CNO); Mixed-effects model: Genotype x Drug: $F_{1,108} = 0.1653$, $p = 0.2027$). (e) Silencing 4-6-week-old abGCs reduces average SWR duration during object trials (Nestin-cre: $n = 5$ (Veh), $n = 7$ (CNO); Nestin-cre:Gi: $n = 9$ (Veh and CNO); Mixed-effects model: Genotype x Drug: $F_{1,106} = 7.9118$, $p = 0.005849$; Tukey's test $p = 0.0094$). (f) Silencing 4-6-week-old abGCs has no effect on SWR peak z-score (Nestin-cre: $n = 5$ (Veh), $n = 7$ (CNO); Nestin-cre:Gi: $n = 9$ (Veh and CNO); Mixed-effects model: Genotype x Drug: $F_{1,111} = 0.0003$, $p = 0.9874$). * $p < 0.05$, bars represent mean + SEM. ns = not significant.



Supplementary Figure 6

4-6-week-old abGC inhibition does not influence theta and gamma power during social interaction.

(a,ai,c,ci,e,ei) Normalized power spectra graphs (Social stimulus/baseline). (b) Silencing 4-6-week-old abGCs has no influence on theta power (Nestin-cre: $n = 7$; Nestin-cre:Gi: $n = 9$; Mixed-effects model: Genotype x Drug: $F_{1,14} = 0.2303$, $p = 0.6387$). (d) Silencing 4-6-week-old abGCs has no influence on low gamma power (Nestin-cre: $n = 7$; Nestin-cre:Gi: $n = 9$; Mixed-effects model: Genotype x Drug: $F_{1,14} = 0.7979$, $p = 0.3868$). (f) Silencing 4-6-week-old abGCs has no influence on mid gamma power (Nestin-cre: $n = 7$; Nestin-cre:Gi: $n = 9$; Mixed-effects model: Genotype x Drug: $F_{1,28} = 0.3702$, $p = 0.5478$). Bars represent mean $\pm$ SEM for a-ai,c-ci,e-ei; + SEM for b,d,e. ns = not significant.

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Article and author information

Blake J. Laham

Princeton Neuroscience Institute, Princeton University, Princeton NJ 08544

Isha R. Gore

Princeton Neuroscience Institute, Princeton University, Princeton NJ 08544

Casey J. Brown

Princeton Neuroscience Institute, Princeton University, Princeton NJ 08544

Elizabeth Gould

Princeton Neuroscience Institute, Princeton University, Princeton NJ 08544

For correspondence: goulde@princeton.edu

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Editors

Reviewing Editor

Laura Colgin

University of Texas at Austin, Austin, United States of America

Senior Editor

Laura Colgin

University of Texas at Austin, Austin, United States of America

Reviewer #2 (Public Review):

Summary:

Laham et al. investigate how the projection from adult born granule cells into CA2 affects the retrieval of social memories at various developmental points. They use chemogenetic manipulations and electrophysiological recordings to test how this projection affects hippocampal network properties during behavior. The study is of relevant interest for the neuroscience community and the results are important for our understanding of how social memories of different nature (remote or immediate) are encoded and supported by the hippocampal circuitry. The behavioral experiments after abGC projections to CA2 are compelling as they show clearly distinct behavioral readout. While the electrophysiological experiments are difficult to interpret without more single cell responses quantifications, they clearly show that more than one region in the hippocampus is involved in the formation of social memories.

<https://doi.org/10.7554/eLife.90600.2.sa1>

Reviewer #3 (Public Review):

Laham et al. present a manuscript investigating the function of adult-born granule cells (abGCs) projecting to the CA2 region of the hippocampus during social memory. It should be noted that no function for the general DG to CA2 projection has been proposed yet. The authors use targeted ablation, chemogenetic silencing and in vivo ephys to demonstrate that the abGCs to CA2 projection is necessary for the retrieval of a remote social memories such as the memory of one's mother. They also use in vivo ephys to show that abGCs are necessary for differential CA2 network activity, including theta-gamma coupling and sharp wave-ripples, in response to novel versus familiar social stimuli.

The question investigated is important since the function of DG to CA2 projection remained elusive a decade after its discovery. Overall, the results are interesting but focused to the social memory of the mother and their description in the manuscript and figures is too cursory. For example, raw interaction times must be shown before their difference. The assumption that mice exhibit social preference between familiar or novel individuals such as mother and non-mother based on social memory formation, consolidation and retrieval should be better explained throughout the manuscript. Thus, when describing the results, the authors should comment on changes in preference and how this can be interpreted as a change in social memory retrieval. Several critical experimental details such as the total time of presentation to the mother and non-mother stimulus mice are also lacking from the manuscript. The in vivo e-phys results are interesting as well but even more succinct with no proposed mechanism as to how abGCs could regulate SWR and PAC in CA2.

The manuscript is well-written with the appropriate references. The choice of behavioral test is somewhat debatable however. It is surprising the authors chose to use a direct presentation test (presentation of the mother and non-mother in alternance) instead of the classical 3-chamber test which is particularly appropriate to investigate social preference. Since the authors focused exclusively on this preference, the 3-chamber test would have been more adequate in my opinion. It would greatly strengthened the results if the authors could repeat a key experiment from their investigation using such test. In addition, the authors only impaired the mother's memory. An additional experiment showing that disruption of the abGCs to CA2 circuit impairs social memory retrieval in general would allow to generalize the

findings to social memories in general. As the manuscript stands, the authors can only conclude as to the importance of this circuit for the memory of the mother. Developmental memory implies the memory of familiar kin as well.

The in vivo ephys section (Figure 3) is interesting but even more minimalistic and it is unclear how abGCs projection to CA2 can contribute to SWR and theta-gamma PAC. In figure 1, the authors suggest that abGCs project preferentially to PV+ neurons in CA2. At minima, the authors should discuss how the abGCs to PV+ neurons to CA2 pyramidal neurons circuit can facilitate SWR and theta-gamma PAC.

Finally, proposing a function for 4-6-week-old abGCs projecting to CA2 begs two questions: What are abGCs doing once they mature further and more generally, what is the function of the DG to CA2 projection? It would be interesting for the authors to comment on these questions in the discussion.

Revision:

The authors have followed my recommendations except for the ones suggesting new experiments. As a result, the clarity of the manuscript and the links between evidence and claims have improved by the message is quite reduced. Many important questions remain open such as: What makes mother's memories so special they require the abGC projection to CA2 unlike other types of social memories? Do abGCs truly connect CA2 PV+ interneurons and how does this connection shape sharp-wave ripples in CA2?

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Author Response

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1:

Summary:

This manuscript provides some valuable findings concerning the hippocampal circuitry and the potential role of adult-born granule cells in an interesting long-term social memory retrieval. The behavior experiments and strategy employed to understand how adult-born granule cells contribute to long-term social discrimination memory are interesting.

We thank the reviewer for the positive evaluation.

I have a few concerns, however with the strength of the evidence presented for some of the experiments. The data presented and the method described is incomplete in describing the connection between cell types in CA2 and the projections from abGCs. Likewise, I worry about the interpretation of the data in Figures 1 and 2 given the employed methodology. I think that the interpretation should be broadened. This second concern does not impact the interest and significance of the findings.

In response to this concern, we have removed the data concerning abGC projections to PCP4+ and PV-GFP+ cell bodies from Figure 1 and have focused this analysis on dendrites. We now provide high magnification images of dendrites and expand on the methodology, results, and interpretations in the manuscript. We also broaden the interpretation throughout the manuscript to address the reviewer's concern.

Strengths:

The behavior experiments are beautifully designed and executed. The experimental strategy is interesting.

We appreciate these positive comments.

Weaknesses:

The interpretation of the results may not be justified given the methods and details provided.

We have addressed this concern by providing more methodological details and broadening our interpretation of the results.

Reviewer #2:

Summary:

Laham et al. investigate how the projection from adult-born granule cells into CA2 affects the retrieval of social memories at various developmental points. They use chemogenetic manipulations and electrophysiological recordings to test how this projection affects hippocampal network properties during behavior. I find the study to be very interesting, the results are important for our understanding of how social memories of different natures (remote or immediate) are encoded and supported by the hippocampal circuitry. I have some points that I added below that I think could help clarify the conclusions:

We appreciate the positive assessment and have addressed the more specific points below.

My major concern with the manuscript was that making the transitions between the different experiments for each result section is not very smooth. Maybe they can discuss a bit in a summary conclusion sentence at the end of each result section why the next set of experiments is the most logical step.

In response, we have added summary conclusion sentences at the end of each result section.

In line 113, the authors say that "the DG is known to influence hippocampal theta-gamma coupling and SWRs". Another recent study Fernandez-Ruiz et al. 2021, examined how various gamma frequencies in the dentate gyrus modulate hippocampal dynamics.

We cite this paper in the revised manuscript.

Having no single cells in the electrophysiological recordings makes it difficult to interpret the ephys part. Perhaps having a discussion on this would help interpret the results. If more SWRs are produced from the CA2 region (perhaps aided by projections from abGC), more CA2 cells that respond to social stimuli (Oliva et al. 2020) would reactivate the memories, therefore making them consolidate faster/stronger. On the other hand, the projections from abGC that the authors see, also target a great deal of PV+ interneurons, which have been shown to pace the SWRs frequency (Stark et al 2014, Gan et al 2017), which further suggests that this projection could be involved in SWRs modulation.

We discuss these possibilities and cite Gan et al 2017, Schlingloff et al., 2014, and Stark et al., 2014 in the revised manuscript.

The authors should cite and discuss Shuo et al., 2022 (A hypothalamic novelty signal modulates hippocampal memory).

We mention Chen et al (A hypothalamic novelty signal modulates hippocampal memory.) in the revised manuscript. “Shuo” is the first name of the first author on this paper, so we believe that this is the same paper to which the reviewer refers.

I think the authors forgot to refer to Fig 3a-f, maybe around lines 163-168.

We thank the reviewer for pointing out this error. In the revised manuscript, we refer to all figure panels. Since Fig 3 is now broken into two figures (Fig 3 and 4), the panel lettering has changed in the revised manuscript.

Are the SWRs counted only during interaction time or throughout the whole behavior session for each condition?

The SWRs are counted throughout the whole behavior session for each condition. This is now stated in the revised manuscript.

Figure 3t shows a shift in the preferred gamma phase within theta cycles as a result of abGC projections to CA2 ablation with CNO, especially during Mother CNO condition. I think this result is worth mentioning in the text.

We now mention this finding in the revised manuscript.

Figure 3u in the legend mention "scale bars = 200um", what does this refer to?

The scale bar refers to that shown in Figure 3b, which is now indicated in the legend.

What exactly is calculated as SWR average integral? Is it a cumulative rate? Please clarify.

The integral measure provides information regarding the average total power of SWR events. It sums z-scored amplitude values from beginning to the end of each SWR envelope, and then takes the average across all summed envelopes. SWR integral has been shown to influence SWR propagation (De Filippo and Schmitz, 2023). This is now described in the text.

Alexander et al 2017, "CA2 neuronal activity controls hippocampal oscillations and social behavior", examined some of the CA2 effects in the hippocampal network after CNO silencing, and the authors should cite it.

Alexander et al., 2018, which we believe is the relevant paper, is now cited in the revised manuscript.

Strengths:

Behavioral experiments after abGC projections to CA2 are compelling as they show clearly distinct behavioral readout.

We thank the reviewer for this positive assessment.

Weaknesses:

Electrophysiological experiments are difficult to interpret without additional quantifications (single-cell responses during interactions etc.)

We have addressed this concern by expanding the interpretation of our results.

Reviewer #3:

Laham et al. present a manuscript investigating the function of adult-born granule cells (abGCs) projecting to the CA2 region of the hippocampus during social memory. It should be noted that no function for the general DG to CA2 projection has been proposed yet. The authors use targeted ablation, chemogenetic silencing, and in vivo ephys to demonstrate that the abGCs to CA2 projection is necessary for the retrieval of remote social memories such as the memory of one's mother. They also use in vivo ephys to show that abGCs are necessary for differential CA2 network activity, including theta-gamma coupling and sharp wave-ripples, in response to novel versus familiar social stimuli.

The question investigated is important since the function of DG to CA2 projection remained elusive a decade after its discovery. Overall, the results are interesting but focused on the social memory of the mother, and their description in the manuscript and figures is too cursory. For example, raw interaction times must be shown before their difference. The assumption that mice exhibit social preference between familiar or novel individuals such as mother and non-mother based on social memory formation, consolidation, and retrieval should be better explained throughout the manuscript. Thus, when describing the results, the authors should comment on changes in preference and how this can be interpreted as a change in social memory retrieval. Several critical experimental details such as the total time of presentation to the mother and non-mother stimulus mice are also lacking in the manuscript. The in vivo e-phys results are interesting as well but even more succinct with no proposed mechanism as to how abGCs could regulate SWR and PAC in CA2.

In response to these comments, we provide raw interaction times in a new Figure (Fig. S1). We also provide more information about the experiments and figures in the revision. We explain the rationale for our behavioral interpretations and discuss proposed mechanisms for how abGCs regulate SWR and PAC.

The manuscript is well-written with the appropriate references. The choice of the behavioral test is somewhat debatable, however. It is surprising that the authors chose to use a direct presentation test (presentation of the mother and non-mother in alternation) instead of the classical 3-chamber test which is particularly appropriate to investigate social preference. Since the authors focused exclusively on this preference, the 3-chamber test would have been more adequate in my opinion. It would greatly strengthen the results if the authors could repeat a key experiment from their investigation using such a test. In addition, the authors only impaired the mother's memory. An additional experiment showing that disruption of the abGCs to CA2 circuit impairs social memory retrieval would allow us to generalize the findings to social memories in general. As the manuscript stands, the authors can only conclude the importance of this circuit for the memory of the mother. Developmental memory implies the memory of familiar kin as well.

We selected the direct social interaction test because it allows for more naturalistic social behaviors than measuring investigation times toward social stimuli located inside wire mesh containers. We also decided to focus our studies on the retrieval of mother memories because these are likely the first social memories to be formed. We emphasize that our results cannot

be generalized to memories of other social stimuli but given studies on recent social memory formation and retrieval in adults that manipulate abGCs and CA2 separately, we feel that it is likely that this circuit is involved in these functions as well. However, we specify throughout the manuscript that our experiments can only tell us about mother memories. We have also changed the title to reflect this.

The in vivo ephys section (Figure 3) is interesting but even more minimalistic and it is unclear how abGCs projection to CA2 can contribute to SWR and theta-gamma PAC. In Figure 1, the authors suggest that abGCs project preferentially to PV+ neurons in CA2. At a minimum, the authors should discuss how the abGCs to PV+ neurons to CA2 pyramidal neurons circuit can facilitate SWR and theta-gamma PAC.

We have divided Figure 3 into two figures (Figures 3 and 4) and revised the electrophysiology section of the results section. In the revised paper, we now discuss how abGC projections to PV+ interneurons may facilitate SWR and PAC.

Finally, proposing a function for 4-6-week-old abGCs projecting to CA2 begs two questions: What are abGCs doing once they mature further, and more generally, what is the function of the DG to CA2 projection? It would be interesting for the authors to comment on these questions in the discussion.

In response to these comments, we discuss possible answers to these interesting questions.

Recommendations for the authors:

Reviewer #1:

Specifically, in Figure 1, for the analysis of the synapses formed between abGCs and CA2 PNS (as identified by PCP4 expression) and CA2 PV+ cells (as identified by cre-dependent AAV-mCherry expression) in PV-cre line. In panels c and d the soma of a CA2 PN cell is shown, as well as the soma of a PV cell is shown. Why was the soma analyzed? What relevance is there for this? It is my understanding that synapses form on dendrites- this would be much more relevant to show, in my opinion. Also, the methods for panels e and f state that the 3R-Tau+ intensity was analyzed only in stratum lucidum. (There was a normalization for the overall 3R-Tau intensity in SL of CA2 that was obtained by dividing the 3R-Tau intensity of corpus callosum). I don't understand then how a comparison of 3RTau intensity could have been done for CA2 PN soma. There are no CA2 PN soma in stratum lucidum. (This is fairly clearly shown in Figure 1a-iii, with the PCP4 staining showing the soma in the somatic layer... not in stratum lucidum). What is being analyzed here?

If the 3R-Tau intensity for dendrites is higher for PV cell dendrites, an example image of dendrites would be very helpful. How was the CA2 PV cell dendrite delimited from the CA2 PN dendrites at 40x magnification for the 3R-Tau intensity? Why were pre-synaptic puncta not examined? Is it possible to determine the post-synaptic target with these methods? This result could be particularly interesting, but I find it very difficult to understand the quantification or the justification behind it. To truly know if a cell is getting a connection, the best method would be to perform whole-cell patch clamp recordings of the post-synaptic target cells and use optogenetics of the abGCs. I understand that perhaps this may be beyond the scope of the paper, but it is a severe limitation for these results.

We have eliminated the cell body measures from Figure 1 and focus instead on the dendrite measures, which we agree are more relevant. We now provide high magnification example images of pyramidal cell (PCP4+) and PV+ interneuron (GFP+) dendrites in Figure 1. We thank

the reviewer for pointing out the error about the stratum lucidum as some of the dendrites analyzed are located in the pyramidal cell layer. In addition, neither PCP4 nor GFP label the full extent of dendrites emanating from CA2 pyramidal cells or PV+ interneurons respectively. We mention this in the revised manuscript because abGC projections to more distal dendrites might show a different pattern than that which was observed for proximal dendrites. We also provide more details about how the dendrites were delimited for the analysis, and mention that these results cannot definitively inform us about whether functional synaptic connections have been formed.

Canulation over CA2 is potentially not specific to CA2 terminals. It would be optimal if the authors had some histology demonstrating specific cannula placement, as these surgeries are really tough to get perfectly centered over CA2. Even if it is perfectly centered, how much would the CNO diffuse into CA3? I think that given the methodology, the authors really need to consider that the behavioral results are not only a result of blocking abGC terminals in CA2 alone. Would it really change much if the abGC terminals are also silenced in CA3a/b as well? The McHugh lab has shown that area CA3 is also playing a role in social memory (Chiang, M.-C., Huang, A. J. Y., Wintzer, M. E., Ohshima, T. & McHugh, T. J. A role for CA3 in social recognition memory. Behav Brain Res 354, 2018). It may be that both areas CA2 and CA3 are important for the phenomenon being demonstrated in Figure 2. I think the impact of the study is just as interesting, as this examination of early social memories is very interesting and nicely done. In fact, areas CA2 and CA3 may be acting together (please see Stöber, T. M., Lehr, A. B., Hafting, T., Kumar, A. & Fyhn, M. Selective neuromodulation and mutual inhibition within the CA3-CA2 system can prioritize sequences for replay. Hippocampus 30, 1228-1238, 2020).

We agree that it is possible that CNO infusions targeted at the CA2 would also influence CA3a/b and have revised the paper to include this possible interpretation. We also cite the suggested paper on CA3 involvement in social memory (Chiang et al., 2018) and the paper on CA2-CA3 interactions (Stöber et al, 2020).

Figure 3 is packed with information, but not communicated in a reasonable way. Much more information and a description of the experimental protocol need to be presented. Furthermore, why are there no example traces for the SWRs recorded? There should be more analysis than just a difference score and frequency. How is j, k, and l analyzed and interpreted? Why no example traces there? Also, the n's seem way too small for Figure 3mr. Are there only 32 or three animals used for some of these conditions? This is insufficient in my opinion to conclude much for a 5-minute interaction.

In response to this concern, we have divided Figure 3 into 2 figures – Figure 3 and Figure 4. In Figure 3, we provide example traces for SWRs, with additional SWR data presented in Figures S3 and S4, including data to complement the difference score data in Figure 3. In Figure 4, we include traces of phase amplitude coupling. We also provide more information in the methods about how the phase amplitude coupling data were analyzed. For Figure 4, we used methods described by Tort et al., 2010 to produce a modulation index, which is a measure of the intensity of coupling between theta phase and gamma amplitude. This method additionally allows for visualization of how gamma amplitude is modified across individual theta phase cycles. Regarding the question about n sizes in the 10-12 week abGC group (Fig. 3), the numbers are lower than in the 4-6 week abGC group because by 6 weeks after the first set of recordings, the electrodes in some of the mice were no longer usable. The n sizes for this specific study are 4-5 per group for Nestin-cre mice; 7-8 for Nestin-cre:Gi. This is now clarified in the figure legend.

The discussion section of this paper does not put these results into a broader context with the field. There are other studies examining abGCs and their roles in novelty and

memory formation (the work from Juna Song's lab, for example). These should be properly mentioned and discussed.

In response, we have added discussion on the roles of abGCs in nonsocial novelty and memory formation and have cited papers from the Song lab.

In the figure legend for Figure 2, there is no specific explanation for panel h. Perhaps the label is missing in the legend.

We thank the reviewer for noting this error and now include a description in the revised manuscript.

Reviewer #2:

Adding more quantifications (single cells, isolating data during interactions versus noninteraction times) would help understand the results better. In the lack of this, adding a more clear rationale (even if only through the presentation of hypotheses) in between the transitions of the different results sections would make the study easier to read.

In response to this comment, we have added transition sentences between results sections to clarify the rationale and make the manuscript easier to understand.

Reviewer #3:

Line 110: "Hippocampal phase-amplitude coupling (PAC) and generation of sharp waveripples (SWRs) have been linked to novel experience, memory consolidation, and retrieval (Colgin, 2015; Fernandez Ruiz et al., 2019; Meier et al., 2020; Joo and Frank, 2018; Vivekananda et al., 2021). The DG is known to influence hippocampal theta-gamma coupling and SWRs (Bott et al, 2016; Meier et al., 2020), yet no studies have examined the influence of abGCs on these oscillatory patterns." This information comes too early in the result section and is somewhat confusing.

In response to this comment, we have moved this information and provided a better description.

Line 118: "we found that mice with normal levels of abGCs can discriminate between their own mother and a novel mother." Be more descriptive of the results (present the raw interaction times with the statistical test to compare them), this is the conclusion.

In response to this comment, we provide the raw interaction times in a new Figure (Fig. S1) and describe the results in more detail.

Line 121: "These effects were not due to changes in physical activity". Be more specific. Did you subject the mice to a specific test? If not, how did you calculate locomotion? The data presented in the supplementary figure 1a only states the % locomotion.

Locomotion was manually scored whenever an animal moved in the testing apparatus. Speed was not recorded. Total locomotion was divided by trial duration to create a % locomotion measure. We have added these details to the methods.

Line 124: "Coinciding with the recovery of adult neurogenesis, GFAP-TK animals regained the ability to discriminate between their mother and a novel mother". Explain how the difference in interaction time can be interpreted as the ability to discriminate. You could also compute the discrimination index used by several other laboratories (difference of interaction normalized by the total interaction time).

In response to this comment, we describe how the difference in interaction time can be interpreted as the ability to discriminate between novel and familiar mice.

Line 133: "Targeted CNO infusion in Nestin-Cre:Gi mice enabled the inhibition of GiDREADD+ abGC axon terminals present in CA2." Provide data or references to support this claim. Injection of a dye of comparable size to CNO would help. Otherwise, mention that nearby CA3a could be affected as well.

We cannot rule out that nearby CA3a was affected by our cannula infusions of CNO into CA2. Furthermore, since dyes likely diffuse at different rates than CNO, we believe that a dye injection would not eliminate this concern completely. Therefore, we have revised the paper to acknowledge the likelihood that the CNO infusion affected parts of CA3 in addition to CA2. We also changed the title to focus more on the CA2 electrophysiological recordings, which we know were obtained only from the CA2.

Line 150: "When reintroduced to the now familiar adult mouse 6 hours later, after the effects of CNO had largely worn off". Provide data or references supporting this claim.

In response, we cite articles that show behavioral effects of CNO DREADD activation are returned to baseline 6 hrs later.

Line 165: "We found that SWR production is increased during social interaction, with more SWRs produced during novel mouse investigation, presumably during encoding social memories, than during familiar mouse investigation, presumably during retrieval of developmental social memories". How does this compare to the results in Oliva et al, Nature 2021?

The Oliva et al 2021 paper recorded CA2 SWRs during home cage and during post-social stimulus exposure periods of sleep. The timing of the study does not coincide with the measures we made, but we cite the paper.

Line 168: "Inhibition of abGCs in the presence of a social stimulus". How does silencing abGC impact CA2 pyramidal neurons' firing rate?

The direct answer to this question is unknown because we did not measure single units, but based on studies done in the CA3, it is likely that firing rate in CA2 would increase.

Line 203: "abGCs possess a time-sensitive ability to support retrieval of developmental social memories." Can you speculate on the function of the cells later on?

In the revised paper, we speculate about the function of abGCs after they mature and no longer support retrieval of developmental social memories.

Line 229: "GFAP-TK mice were group housed by genotype". Why not housed them with CD1 littermates?

We housed these mice according to genotype to avoid having mice with different levels of abGCs (GFAP-TK + VGCV and CD1 + VGCV) living together in social groups. We did this to avoid potential differences that might emerge in social behavior.

Line 237: "Adult TK, Nestin-cre, and Nestin-cre:Gi offspring underwent a social interaction test in which they directly interacted with the mother". Specify how long was the social interaction time.

In the revised manuscript, we specify that mice interacted with each social stimulus for 5 minutes.

Line 240: "After a 1-hour delay spent in the home cage". Were the mice single-housed or with their littermates during this delay?

In the revised manuscript, we indicate that mice were put back into the home cage with their cagemates during the 1 hr delay period.

Line 241: "The order of stimulus exposure was counterbalanced in all tests." Can you show some data to confirm that the order of presentation did not impair the interaction? Have you considered using your own version of the classical 3-chamber test in order to assess directly the preference for one or the other female mouse?

Our data suggest that the order of testing is not responsible for the observed results. Across all experimental groups without an abGC manipulation (i.e., all direct social interaction assays excluding VGCV+ GFAP-TK trials and CNO+ Nestin-cre:Gi trials), ~84.4% of animals demonstrate a social preference for the novel mother over the mother (CD1 + GFAP-TK VGCV-cohort: 28/33; CD1 VGCV+ cohort: 17/17; CD1 and TK recovery cohort: 24/31; Nestin-cre and Nestin-cre:Gi 4-6-week-old abGC cohort: 77/95; 10-12-week-old abGC cohort: 49/55; Total = 195/231 mice with an investigation preference for the novel mother). If stimulus presentation order were to bias social investigation preference toward the first stimulus presented, we would expect the percentage of animals demonstrating a social preference for each stimulus to be around 50%, as roughly half the animals were first exposed to the mother with the other half first exposed to the novel mother. The social novelty preference percentage reported above is comparable to percentages we observe in our lab's novel to familiar social interaction experiments, in which all animals are first exposed to a novel conspecific. We have yet to conduct experiments testing adults using the modified 3-chamber assay described in Laham et al., 2021.

Statistics: The statistical tests used throughout the paper are appropriate but their description is too cursory. Please provide F values and specify the name of the tests used in the figure legends before giving the exact p values.